RingMo: A Remote Sensing Foundation Model With Masked Image Modeling

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***Abstract*— Deep learning approaches have contributed to the rapid development of remote sensing (RS) image interpretation. The most widely used training paradigm is to use ImageNet pretrained models to process RS data for specified tasks. However, there are issues such as domain gap between natural and RS scenes and the poor generalization capacity of RS models. It makes sense to develop a foundation model with general RS feature representation. Since a large amount of unlabeled data is available, the self-supervised method has more development significance than the fully supervised method in RS. However, most of the current self-supervised methods use contrastive learning, whose performance is sensitive to data augmentation, additional information, and selection of positive and negative pairs. In this article, we leverage the benefits of generative self-supervised learning (SSL) for RS images and propose an RS foundation model framework called RingMo, which consists of two parts. First, a large-scale dataset is constructed by collecting two million RS images from satellite and aerial platforms, covering multiple scenes and objects around the world. Second, we propose an RS foundation model training method designed for dense and small objects in complicated RS scenes. We show that the foundation model trained on our dataset with RingMo method achieves state-of-the-art (SOTA) on eight datasets across four downstream tasks, demonstrating the effectiveness of the proposed framework. Through in-depth exploration, we believe it is time for RS researchers to embrace generative SSL and leverage its general representation capabilities to speed up the development of RS applications.**

***Index Terms*— Foundation model, masked image modeling (MIM), pretraining, remote sensing (RS), self-supervised, Vision Transformer (ViT).**

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I. Introduction

In RECENT years, the number of satellite launches has exploded [1]–[6]. According to the report released by the Union of Concerned Scientists,[1] there are more than 1000 Earth observation satellites in orbit as of December 2021. The potential to acquire huge amounts of remote sensing (RS) satellite data comes with it. For example, the Zhuhai-1 group satellite can download nearly 20TB of data every day [7]. Massive RS data need to be processed and analyzed to play a role in application tasks. Benefiting from the deep learning, many tasks using RS data have made remarkable progress [8]–[17]. At present, the general paradigm of RS image interpretation is to load the ImageNet pretrained weights, and then use RS data and label information in the specified task for model training [18]–[21]. Although these methods perform well for specific targets in specific tasks, there still exist the following issues. 1) As shown in Fig. 1, due to the differences between natural and RS scenes, ImageNet pretrained weights have domain bias, which impacts the performance of RS models to some extent. 2) The model's generalization capacity is relatively weak since the restricted RS training data can only supply a small number of scenes and objects, making it difficult to adapt to other tasks. 3) To scale to more targets and tasks, researchers need to continuously annotate data, which reduces the efficiency of practical applications. To sum up, using massive RS data and designing a foundation model with the general RS feature representation will be expected to solve the above problems.

[1] https://www.ucsusa.org/resources/satellite-database

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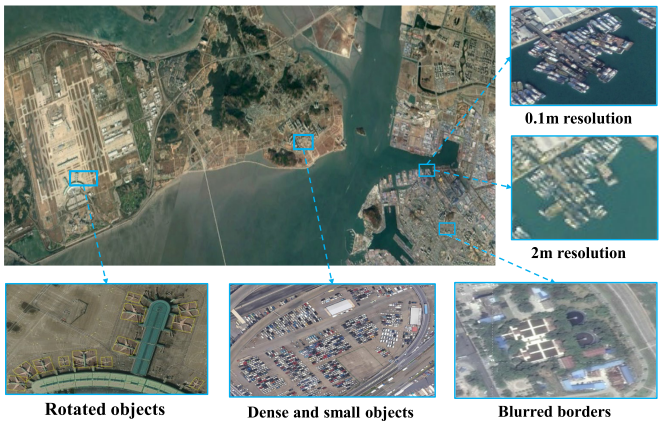

Fig. 1. Some examples of RS images. RS image has crowded scenes and multiple resolutions, containing objects with diverse angles, various sizes, and blurred borders.

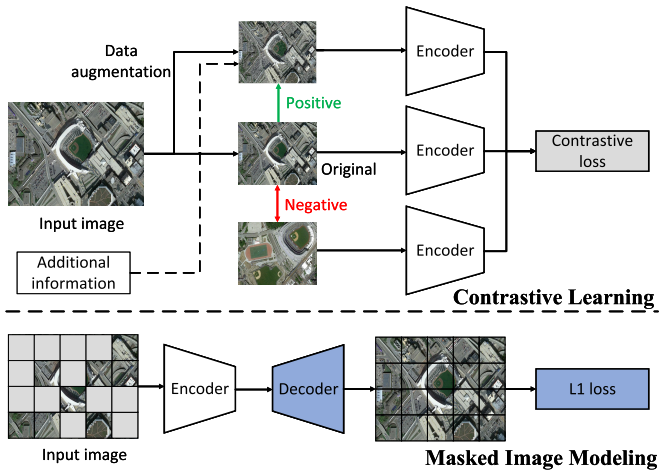

Fig. 2. Brief introduction of the contrastive learning methods and MIM method for the RS foundation model. Contrastive learning methods pull the positive pairs closer and push the negative pairs away. MIM method first masks a proportion of the image, and then a pipeline is used to reconstruct masked parts.

The existing RS foundation model training methods consist of two mainstreams, supervised and self-supervised learning (SSL). For supervised learning [22], a large collection of labeled data is necessary to train the foundation model. Although these methods work well, obtaining labeled data requires a lot of time from RS experts. Because of the enormous gap between data labeling and data acquisition speed, a large amount of RS data is unable to be used. Self-supervised methods [23]–[25] can leverage a significant quantity of unlabeled data and are the mainstream methods for the RS foundation model. Contrastive learning [26], [27] dominates the field of RS self-supervised methods. As shown in Fig. 2, the core idea of contrastive learning is to close the feature distance of similar samples. Since there is no label, researchers [23], [28] create positive pairs by data augmentation of the same image, while other images serve as negative pairs for this image. Furthermore, some researchers [24], [25], [29], [30] introduce additional information specific to RS (geographic information, time-series data, audio data, etc.) through careful search or matching and encourage models to learn similarities with additional information in addition to data augmentation. Contrastive learning methods achieve great performance, but they default to different images as negative pairs, and different images in RS datasets frequently contain instances of the same class.

In the natural scene, the generative self-supervised method has steadily gained popularity. Masked image modeling (MIM) [31], [32] is a kind of SSL method applying the generative model. It aims to reconstruct the masked pixels and learn the general feature representation among data distribution. On one hand, it avoids introducing additional information, making it easier to use large amounts of data. The foundation model can gain more adaptability to different scenes as more data are used for modeling. On the other hand, its goal is to reconstruct pixel-level signals from the original image. Without any data augmentation and negative pairs, the model can also derive superior feature representation. As a result, generative self-supervised methods are more suitable for developing RS foundation model.

Currently, most MIM methods are trained based on natural images [31], [32]. Compared with natural scenes, RS images have the following difficulties.

1) *Large Resolution and Orientation Ranges:* Influenced by RS sensors, images have multiple spatial resolutions. Furthermore, different from natural images whose instances usually have fixed orientation due to gravity, objects in RS images have a large angular distribution range from a bird's eye view. Therefore, due to the diversity of scales and angles, the same object has different characteristics in different RS images.
2) *Many Dense and Small Objects:* A large proportion of natural images contain a few objects. For example, the ImageNet dataset contains less than three object instances per image on average [33]. As shown in Fig. 1, RS images are usually large and wide, covering hundreds of kilometers. There are many small objects in the RS images, and they are usually densely distributed, which affects object-level interpretation accuracy to a certain extent.
3) *Complex Background:* Since the RS image contains a relatively large scene, in addition to the object of interest, the image also contains a lot of background information, resulting in a low signal-to-noise ratio of the image. The boundary and background of objects are blurred, which interferes with object classification. Moreover, RS images are easily interfered by external factors such as weather, light, clouds, and fog, which affect the imaging quality.

Based on the above differences, generative self-supervised models trained with natural images perform not very well on RS scenes. Therefore, it is necessary to design a foundation model suitable for RS data.

In this article, we propose a **r**emote sens**ing** foundation **mo**del framework **RingMo**, which consists of two parts: a large-scale RS dataset and a foundation model training method suitable for multitask interpretation. With more RS

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data, the foundation model can learn more general feature representations. To cover RS data with complex backgrounds and numerous objects, we collect a dataset of two million images under unsupervised conditions. The RS images, which range in resolution from 0.1 to 30 m and cover multiple scenes and objects across six continents, are captured from both satellite and aerial platforms. Based on this dataset, we design the RingMo training method to apply the MIM method to train an RS foundation model. The pipeline consists of a relatively heavy encoder for extracting latent feature representations and a decoder for outputting reconstruction results. The existing random masking strategy tends to lose dense and small objects in RS images. To solve this problem, we design a patch incomplete mask (PIMask) strategy according to the properties of RS scenes and objects. The RS objects are preserved in complicated RS scenes by adopting partial incomplete masks while keeping the overall mask ratio.

We use RingMo to train on our dataset to obtain the foundation model, and the trained feature representation can be applied to a wide range of RS downstream tasks. We evaluate our method on four RS tasks. We use UCM [34], AID [35], NWPU-RESISC45 [36] for scene recognition experiments, FAIR1M [4] and DIOR [5] for object detection experiments, iSAID [37] and ISPRS Postdam[2] for semantic segmentation experiments, and LEVIR-CD [38] for change detection experiments. The experimental results demonstrate that our foundation model can achieve state-of-the-art (SOTA) performance on eight datasets. In addition, as our framework does not need to manually annotate any labels, we also discuss the impact of different volumes of RS data on foundation model training, and increasing the number of RS data can obtain better experimental results.

This article's contribution can be summarized into fourfolds.

1) We propose the first generative self-supervised foundation model framework (RingMo) in the field of RS. This framework exploits vast RS data to obtain general feature representation and increase the accuracy of various RS interpretation tasks.
2) To enhance the processing ability of the foundation model for RS data, we design a self-supervised method based on the properties of RS images and improve the situation where the previous mask strategies may ignore dense and small objects in complicated RS scenes.
3) Without any human supervision, we collect an RS dataset of two million images, which are captured from satellite and aerial platforms, covering diverse objects and scenes on six continents. Such dataset containing a large and diverse range of RS images improves the adaptability of the foundation model to different scenes.
4) After deriving the foundation model using the RingMo training method on the collected dataset, we fine-tune it on four typical RS tasks. The experiments show that our method achieves SOTA in eight downstream datasets and verifies the effectiveness and generalization of our RS foundation model on various applications.

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

II. Related Works

In this section, we mainly introduce the foundation model relevant works in the field of RS and the generative self-supervised models in the field of computer vision (CV).

A. Self-Supervised Methods in RS Scene

Similar to SSL in the natural scene, the RS self-supervised models typically use huge amounts of unlabeled RS data to mine representation information that can be transferred to a variety of downstream tasks, including classification, segmentation, detection, etc. At present, the SSL methods in the field of RS are basically contrastive learning methods.

Partial research works based on the existing popular contrastive learning fameworks (e.g., MoCo [42]–[44], SimCLR [26], [27], and CMC [45]) to produce positive and negative samples through data augmentation (e.g., random crop, color distortion, and Gaussian blur). Zheng *et al.* [46] combine recent contrastive learning methods such as MoCo and cross-level nstance-group discrimination with geometric augmentations to improve the recognition of rare wildlife recognition in UAV images. While Stojnic and Risojevic [47] leverage CMC on different RS datasets. They show that for downstream task classification in RS images, using self-supervised pretraining on RS data can achieve better results than using supervised pretraining on natural scene images. Similarly, motivated by CMC, Cha *et al.* [48] propose multimodal representation for synthetic aperture radar (SAR) semantic segmentation by jointly using electro-optical (EO) imagery, SAR imagery, and a label mask. Based on the SimCLR framework, Jung *et al.* [49] propose contrastive SSL for RS with smoothed representation by leveraging multiple-input images and averaging their representations.

There are other works that introduce some additional information such as spatio-temporal information and geographical knowledge to design positive and negative samples for contrastive learning. For example, Mañas *et al.* [29] propose seasonal contrast (SeCo) to take advantage of temporal information to acquire image pairs from the same location at different time points, which provides a natural enhancement to assist contrastive SSL. Ayush *et al.* [25] exploit spatially aligned images over time to build temporal positive pairs to design pretext tasks in contrastive learning. Akiva *et al.* [28] propose a self-supervised approach based on material and texture representation. Li *et al.* [24] present a geographical knowledge-driven representation learning technique for RS images (GeoKR) and construct a large-scale pretraining dataset Levir-KR. Dong *et al.* [50] leverage temporal prediction to deal with the change detection task. While Chen *et al.* [51] construct a dataset of bitemporal images with semantic masks and propose a semantic decoupled representation for change detection.

B. Generative Self-Supervised Methods in CV

As a branch of SSL, generative self-supervised models have been widely studied in CV. These methods use the input data itself as supervision to learn representations and apply

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TABLE I
STATISTICAL INFORMATION OF DATASETS FOR RS FOUNDATION MODEL TRAINING

Dataset	Source	The Number of Images	Image Sizes	Spatial Resolution
MATTER [28]	Sentinel-2	$14,857$	$1,096$	-
Sen12MS [39]	Sentinel-1/2	$12,228$ pairs	-	-
SeCo [29]	Google Earth	1 million	≈ 600	$10-60m$
Million-AID [40]	GeoEye, WorldView, QuickBird, IKONOS, SPOT, Landsat serial satellites	1 million	$256/512$	$0.5-153m$
fMoW [41]	QuickBird, WorldView	$1,047,691$	224	-
Levir-KR [24]	Gaofen	$1,431,950$	256	$0.8-16m$
Our dataset	Gaofen, JL-1, GeoEye, WorldView, QuickBird, SPOT, IKONOS, Landsat serial satellites	$2,096,640$	448	$0.1-30m$

them to various downstream tasks of the vision field. The recent generation models mainly consist of auto-regressive (AR) models, flow-based models, and auto-encoding (AE) models.

AR models can be regarded as "Bayesian network structures," which predict new data samples based on previous data. PixelRNN [52] and PixelCNN [53] model images pixel by pixel with RNN and CNN, respectively. Under the assumption that there is a dependency relationship between pixels of the input image, they generate lower (right) pixels by adjusting upper (left) pixels. Flow-based models aim to estimate complex high-dimensional densities from data. For instance, NICE [54], RealNVP [55], and Glow [56] design affine transformations to parameterize data distribution.

AE models typically involve an encoder that generates latent representations and a decoder that reconstructs inputs, which are more popular due to their flexibility. Denoising autoencoders (DAEs) [57] propose to recover the original signal from the corrupted signal, making the representation robust to noise. There are a series of methods based on DAE with different corruptions, e.g., masking pixels [58]–[60] or removing color channels [61].

MIM can be regarded as a DAE model, which learns to reconstruct the original uncorrupted image from the masked input image. Context encoder [59] encodes images that are to be patched by convolution network and outputs completed images. Benefiting from the success of Vision Transformer (ViT) [62] that adapts transformers that were previously generalized for NLP to vision tasks, most of the recent AE methods in CV are conducted with Transformers. iGPT [60] resizes raw images to a low resolution and converts them directly into a 1-D sequence as input. Next pixels or masked pixels can be predicted in sequences of pixels. Following BERT developed in NLP, BEiT [63] proposes to transform the original image into visual tokens and recover the original discrete token based on corrupted image patches by randomly masking some image patches. Most recently, He *et al.* [31] proposed MAE that masks random patches of the input image and reconstructs the missing pixels. SimMIM [32] designs a simple MIM method and replaces masked patches with learnable mask token vectors. Masked and visible tokens are fed together as input to predict the original pixel values of masked patches by the encoder–decoder, which allows SimMIM to be applied to Swin Transformer as well. However, to our best knowledge, there is little exploration of MIM-based SSL for the RS foundation model. Inspired by the remarkable results of generative self-supervised models in CV, our work is devoted to exploring such methods in RS field.

III. RINGMO FRAMEWORK

This article proposes an RS foundation model developing framework RingMo, which aims to obtain the foundation models trained from the collected dataset in a generative SSL manner. In this section, we first introduce our proposed dataset in the aspect of statistical information, capture sources, and scene distribution. Then we give a brief pipeline description of SSL applying the RingMo training method. Among them, masking strategy, model architecture, and reconstruction target will be described in detail.

A. Dataset Description

In this section, we collect a large-scale dataset for the SSL task in the RS field. This dataset comprehensively covers various RS scenes, possessing the characteristics of multisource, multitemporal, and multi-instance. The details of the dataset will be indicated in the subsequent content from three aspects: statistical information, capture sources, and scene distribution.

1) Statistical Information: Some of the images in the dataset are selected from public datasets in the RS field, such as [4], [5], [37], [40], [64], [65], and the other images are obtained from China's Gaofen-2 satellite. Then these images captured by variant platforms are cropped into 448 $\times$ 448 pixels, and 2 096 640 images with multiple resolutions and scenes are obtained. The complex characteristics of the dataset can enable the foundation model to learn a more robust feature representation that is more suitable and specific for downstream tasks in the RS field. The description of our dataset and the comparison with other datasets used for foundation model training are shown in Table I. Our dataset has the largest number of images, and the resolution of the dataset varies in a very large range which is from 0.3 to 30 m.

2) Capture Sources: The used images are captured from different sensors having diverse imaging parameters. Compared with other existing datasets for the RS foundation model training task, the images are captured from both the aerial and satellite platforms: Satellite JL-1, Satellite GF-2, Ganfen satellites, GeoEye, WorldView, QuickBird, IKONOS, SPOT, etc. There exist discrepancies between the aerial images and the satellite images to make foundation models behave better.

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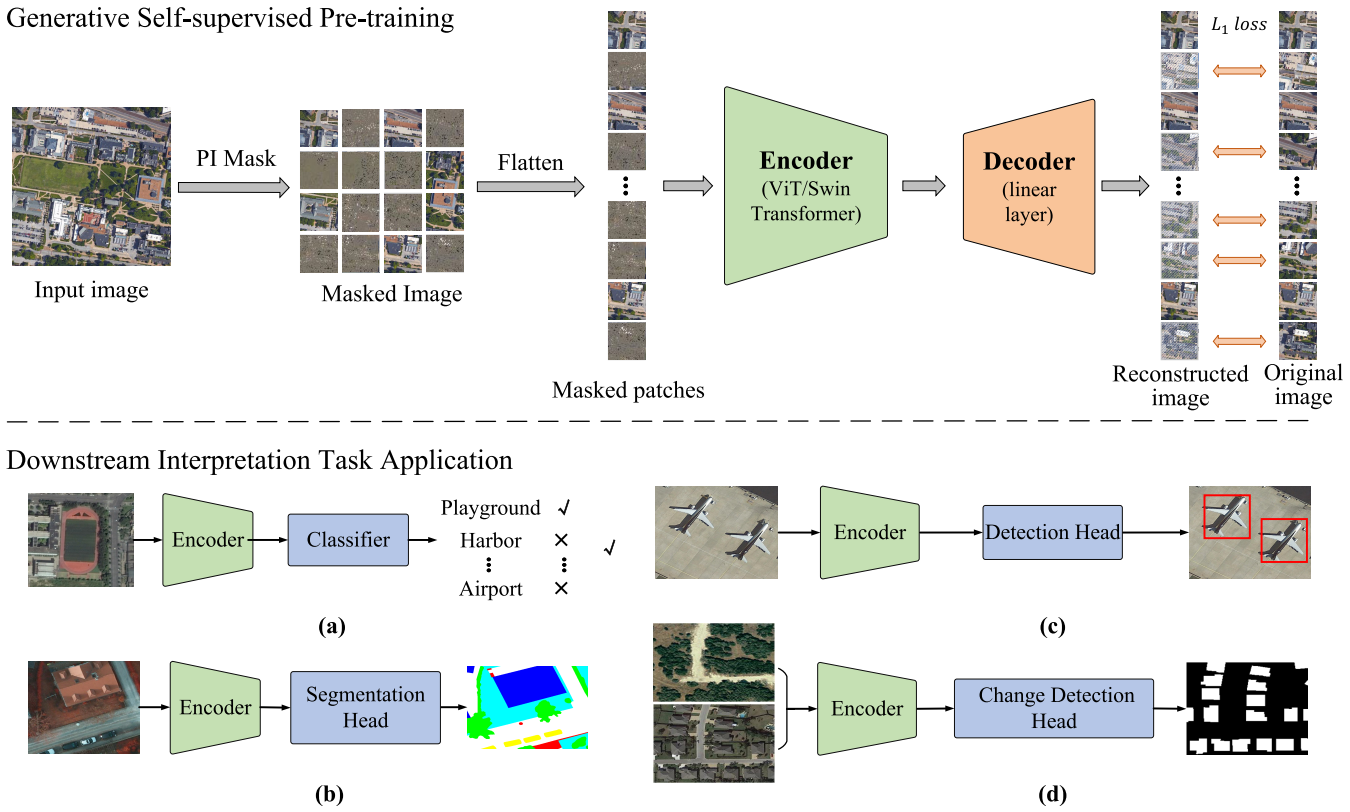

Fig. 3. Illustration of our method for training RS foundation model and fine-tuning on downstream interpretation tasks. Given an optical RS image, the PIMask strategy is used to mask the selected area, and then the flattened patches are sent to the encoder and decoder for the reconstruction of the masked patches. The reconstructed results are calculated $L1$ loss with the original image pixels' value. After pretraining, we attach various task-specific heads to the encoder and fine-tune the overall parameters. (a) Scene classification. (b) Semantic segmentation. (c) Object detection. (d) Change detection.

The shooting angle of the satellite images is often vertical, while the shooting angle of the aerial images contains sloped situation. The image quality of aerial images is usually higher than that of satellite images, yet the satellite images provide greater coverage than aerial images.

3) Scene Distribution: The used images cover different scenes in six continents: Asia, Europe, North America, South America, Africa, and Oceania. Images taken in different seasons and times are also included. The illumination level is also different between images, such as the light intensity, the difference caused by the shooting time, shadows caused by shading, etc. All the mentioned situations introduce intraclass variation and interclass similarity in the downstream tasks, which pose a severe challenge to obtain a highly generalizable model. Therefore, images with as many different characteristics as possible are used to perform the pretraining process to learn a better unified feature representation.

B. Foundation Model Training Method

The proposed RingMo training method learns RS representations through generative SSL. This modeling is a typical AE approach, which reconstructs from partial observations of the original signal. To avoid losing the feature information of small objects, we design the PIMask strategy. Given an input image, PIMask implements region selection and mask generation. Like other autoencoders, our method has an encoder that extracts latent representations of masked images, which are then used to reconstruct the raw signals for masked regions. The learned encoder should be useful for various optical RS downstream tasks. In this work, we mainly take into account two classical visual Transformer architectures: ViT [62] and Swin Transformer [66]. The reconstruction target specifies the form of the original signal to be predicted, and $L1$ regression loss is used for calculating the difference between the reconstructed results and the pixel value. The proposed framework architecture is shown in Fig. 3.

In Sections III-B1–III-B3, we will systematically introduce each component.

1) PIMask Strategy: The masking strategy commonly used in most of MIM methods is random masking, as shown in Fig. 4. A certain proportion of image patches are randomly selected and then totally masked. This method is useful in natural images. However, there are some problems in the application of RS images. The special imaging mechanism leads to more complex background and small-scale objects. It is easy to ignore many whole small objects with the random mask strategy. Just like the right red patch in Fig. 4 shows, the random mask strategy complete loses the small target information in masked patches, which affects the foundation model to reconstruct the small objects and increase the difficulty of image reconstruction. Therefore, we design a new mask strategy named PIMask to address this problem.

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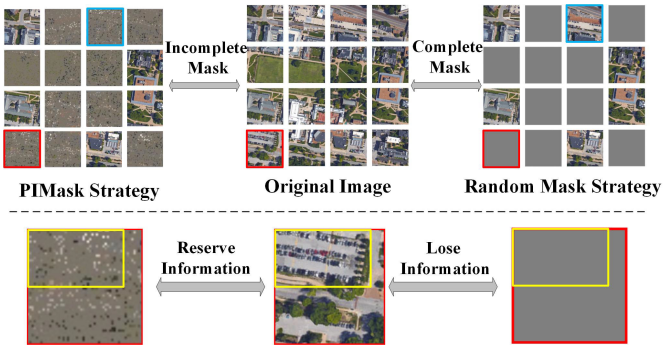

Fig. 4. Visual comparison of mask strategies between the RingMo MIM method and previous MIM method. Using PIMask strategy can reserve partial area of small objects, while random mask strategy loses them. We recommend to enlarge image to see better details.

As shown in the left red patch in Fig. 4, instead of masking image patches completely, we randomly reserve some pixels in masked patches. With this mask strategy, some pixel information of small targets can be reserved effectively. Just like the blue patches in Fig. 4 show, we increase the masked patches number to keep the total mask ratio unchanged.

Besides, to make better use of these reserved pixels, multilayer convolution is adopted to realize patch embedding. Some relevant researchers have experimentally proved that adding early convolution layers into ViT can help model learn image feature better [67]. Specifically, in the process of convolution, we make convolution kernel calculate only inside each patch, which cannot break the mask constraint of the model. Different from the traditional embedding results, all the tokens after multilayer convolution have feature information, which further improve the learning efficiency of the encoder.

The mathematical calculation process of the PIMask strategy is shown as follows. First, we slice an image into a collection of many nonoverlapping patches

$$\begin{aligned} m_{\text{tag}} &= \text{RTG}(\alpha, \text{len}(C_{\text{patches}})) \\ &= \text{RTG}(\alpha, \text{len}(\{\text{patch}_i\}, (i = 1, 2, 3, \ldots, n))). \end{aligned} \quad (1)$$

As shown in Formula (1), we take the proportion of masked patches α and the total number of patches $\text{len}(C_{\text{patches}})$ as input and determine the mask label of each patch through randomly label generator (RTG) function. Therefore, the output m_{tag} is a 1-D vector which includes all mask labels of the patches

$$\text{PIMask}_i = \begin{cases} \text{RTG}(\beta, \dim(\text{patch}_i)), & m_{\text{tag}}[i] = 1 \\ \text{RTG}(0, \dim(\text{patch}_i)), & m_{\text{tag}}[i] = 0 \end{cases} \quad (2)$$

$$\text{patch}^i_{\text{mask}} = \text{patch}_i \times (I - \text{PIMask}_i). \quad (3)$$

After obtaining the mask label $m_{\text{tag}}[i]$ of patch_i, we can determine whether the patch is partially masked. As shown in (2), we take the mask inside ratio β and the dimension of the patch $\dim(\text{patch}_i)$ as input and generate the mask tensor PIMask_i of each patch through RTG function. As shown in (3), we directly multiply patch_i with $I - \text{PIMask}_i$, where I represents the unit matrix and $\times$ represents the element-by-element multiplication.

2) Model Architecture: The main components of the model include the encoder and the decoder, where the encoder includes two typical visual Transformers: ViT [62] and Swin Transformer [66]. We feed both visible and masked patches along with projections of positional embeddings into the encoder. Specifically, ViT receives the transformed sequence of token embeddings as input. Similar to BERT [68], ViT prepares a learnable class embedding for the embedded token sequence in advance, which aims to represent the global class at the output of the Transformer encoder. ViT uses standard learnable 1-D positional embeddings. The transformer encoder includes alternating multilayer perceptron and multihead self-attention layers. A hierarchical module and residual connections are used before each part. The vision-specific inductive bias in ViT is less than that in convolutional neural networks (CNNs).

For the Swin Transformer, it first crops the image into patches without overlaps. Each patch is viewed as a "token" whose features are set as the aggregation of the original pixel information. The projection layer is used to tokenize each patch to a random dimension (indicated as C). Several specified Swin Transformer blocks are used on patch tokens. Together, these blocks generate a hierarchical representation with the same resolution as classical CNNs. The Swin Transformer is constructed by changing the standard multihead self-attention in the Transformer block into the moving window-based module, and other layers remain unchanged. Therefore, the Swin Transformer can conveniently replace the backbone network in previous approaches for several tasks.

As mentioned above, we choose two classical ViT architectures as the encoder, respectively. The decoder is just used in pretraining for image reconstruction. In this article, we experiment with linear layers, which is shallower and narrower than the encoder.

3) Reconstruction Target: Since pixel values are continuous in optical space, our method reconstructs the input by regressing the original pixels that predict masked regions. The loss function calculates the L_1 distance between the reconstructed image and the raw image

$$L = \frac{1}{\Omega(\mathbf{x}_M)} \|\mathbf{y}_M - \mathbf{x}_M\|_1 \quad (4)$$

where $\mathbf{x}, \mathbf{y} \in R^{3HW \times 1}$ indicate the raw pixel values and reconstructed pixel values, respectively; Ω is the number of elements; M represents the set of masked pixels. Similar to other MIM methods [32], we only compute the loss for masked regions.

IV. Experiments

A. RS Foundation Model

We use the RingMo training method to train the RS foundation models with our collected dataset. To verify the effect of RS data on the foundation model, we use large-scale RS data to train foundation models with a random masking strategy, and we also evaluate the impact of different data amounts and training epochs. Furthermore, to validate our proposed innovative method, we use PIMask for training the foundation model. Subsequent experiments show that the proposed RS foundation model framework can be effectively implemented

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Fig. 5. Examples of visualized results for our RingMo method with the PIMask strategy. For each batch, (Left to right) original image, masked image, reconstructed image, and combination image (pixels of masked patches come from reconstructed image and pixels of unmasked patches come from original image), respectively.

on RS images, and more importantly, it can effectively improve the performance on various downstream interpretation tasks without complex operations.

All the experiments for the foundation models are implemented by the Pytorch framework [69] with NVIDIA Tesla V100 GPUs. We adopt Swin-B and ViT-B as the encoder backbone for pretraining. The input image size of ViT-B is 224, and the input image size of Swin-B is 192. The patches are generated from the dataset through random position cropping. We set the batch size as 128 per GPU and use an AdamW optimizer with a momentum of 0.9 and a weight decay of 0.05. The hyperparameters of the optimizer are set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. The base learning rate (LR) is set to $2e^{-4}$. We warm up the model for ten epochs and pretrain the model for 200 epochs. Our foundation model training also supports the MindSpore framework,[3] and the codes and foundation models' weights will be released.

To verify the reconstruction performance, we visualize the reconstructed images of the foundation model with the PIMask strategy in Fig. 5. It shows that the generative self-supervised model pretrained on the RS dataset basically reconstructs the masked areas according to the context information of the reserved patches. In addition, as shown in Fig. 5, these reconstruction results differ slightly from the original images, showing that our model reconstructs images truly based on the unmasked regions instead of simply memorizing original images and possesses generalization ability.

After pretraining the foundation model, we attach task-specific heads to the pretrained backbone and fine-tune the parameters of downstream tasks, as shown in Fig. 3. We verify our foundation model on common tasks in RS image interpretation, such as scene classification, object detection, semantic segmentation, and change detection. The experimental results validate the effectiveness of our generative self-supervised model pretrained on the constructed RS dataset. With the proposed RingMo framework, it is straightforward to exploit the pretraining-then-fine-tuning paradigm in other RS interpretation tasks.

[3]https://gitee.com/mindspore/ringmo-framework

B. RS Scene Classification

For the scene classification task, we use a linear classifier as the task head. Concretely, average pooling is used to fuse representations and pass global information to the classifier. Class probabilities are calculated as softmax values. The likelihood of labeled data is maximized by optimizing the parameters of the RS foundation model and softmax classifiers.

1) Dataset Introduction: Three classic RS scene classification datasets are used to comprehensively evaluate the role of the proposed RS foundation model and the ability of the above backbone networks.

1) *AID [35]:* This challenging dataset is crafted from multisensor data collected on Google Earth. The spatial resolution varies from 8 to 0.5 m, and the size is 600×600 pixels. The AID dataset consists of 10 000 images, 30 categories in total. Each category possesses a number ranging from 220 to 400 images. Smaller interclass variance, larger intraclass variance, and imbalanced distribution of classes make AID quite challenging to current scene classification methods.
2) *NWPU-RESISC45 [36]:* This dataset is a fairly large-scale benchmark obtained from Google Earth. Covering a large number of regions around the world, it contains a total of 31 500 images. There are 45 categories, with 700 images in each. The spatial resolution of each RS image ranges from about 0.2 to 30 m and is 256×256 pixels in size.

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TABLE II
COMPARISONS WITH THE SOTA MODELS ON THREE SCENE CLASSIFICATION DATASETS UNDER DIFFERENT SETTINGS. THE OA IS ADOPTED AS THE EVALUATION METRIC

Method	Publication Year	AID TR=20%	AID TR=50%	NWPU-RESISC45 TR=10%	NWPU-RESISC45 TR=20%	UCM TR=80%
DCNN [70]	2018	90.82	96.89	89.22	91.89	98.93
CBAM [71]	2018	94.66	96.90	92.10	94.26	99.04
MSCP [72]	2018	92.21	96.56	88.07	90.81	98.40
Two-Stream Fusion [73]	2018	92.32	94.58	80.22	83.16	98.02
RTN [74]	2018	92.44	–	89.90	92.71	98.96
GCFs+LOFs [75]	2018	92.48	96.85	–	–	99.00
Hydra [76]	2019	–	–	92.44	94.51	–
CapsNet [77]	2019	91.63	94.74	85.08	89.18	98.81
ARCNet [78]	2019	88.75	93.10	–	–	99.12
SCCov [79]	2019	93.12	96.10	89.30	92.10	99.05
KFBNet [80]	2020	95.50	97.40	93.08	95.11	99.88
GBNet [81]	2020	92.20	95.48	–	–	98.57
MG-CAP [82]	2020	93.34	96.12	90.83	92.95	99.00
EAM [83]	2021	94.26	97.06	91.91	94.29	99.21
EAM [83]	2021	93.64	96.62	90.87	93.51	98.98
F^2BRBM [84]	2021	96.05	96.97	92.74	94.87	99.58
MBLANet [85]	2021	95.60	97.14	92.32	94.66	99.64
GRMANet [86]	2021	95.43	97.39	93.19	94.72	99.19
IDCCP [87]	2021	94.80	96.95	91.55	93.76	99.05
MSANet [88]	2021	93.53	96.01	90.38	93.52	98.96
CTNet [89]	2021	96.25	97.70	93.90	95.40	–
LSENet [90]	2021	94.41	96.36	92.23	93.34	99.78
DFAGCN [91]	2021	–	94.88	–	89.29	98.48
MGML-FENet [92]	2021	96.45	98.60	92.91	95.39	99.86
ESD-MBENet-v1 [93]	2021	96.20	98.85*	93.24	95.50	99.86
ESD-MBENet-v2 [93]	2021	96.39	98.40	93.05	95.36	99.81
SeCo-ResNet-50 [29]	2021	93.47	95.99	89.64	92.91	97.86
RSP-ViTAEv2-S [22]	2022	96.91*	98.22	94.41*	95.60	99.90*
ISP(ViT)[62]	2020	96.24	97.95	93.17	95.23	99.46
ISSP(ViT)[32]	2021	95.82	97.98	93.04	94.87	99.05
RingMo(ViT-100W-200E)	–	96.27	98.13	93.01	95.28	99.31
RingMo(ViT-150W-200E)	–	96.52	98.36	93.38	95.32	99.48
RingMo(ViT-200W-200E)	–	96.54	**98.38**	93.46	95.35	99.46
ISP(Swin)[66]	2021	96.18	98.03	93.12	94.97	99.27
ISSP(Swin)[32]	2021	95.89	97.95	93.14	95.05	98.83
RingMo(Swin-100W-200E)	–	96.17	98.17	93.07	94.97	99.13
RingMo(Swin-150W-100E)	–	96.09	98.21	92.94	95.02	99.06
RingMo(Swin-150W-200E)	–	96.42	98.32	93.19	95.14	99.28
RingMo(Swin-150W-200E†)	–	96.59	98.16	93.33	95.39	99.29
RingMo(Swin-200W-200E)	–	**96.90**	98.34	**94.25**	**95.67***	**99.84**

3) *UCM [34]:* As a typical dataset in the scene classification task, it covers 2100 RS images. They are all from the USGS National Map, with a resolution of 1 ft. Each image is 256 pixels wide and high. The images are evenly distributed among 21 categories, i.e., 100 images per category. The images are collected from various urban areas in the United States and are extremely challenging.

2) Detailed Experimental Settings: All the experiments are implemented on the PyTorch framework [69]. The hardware uses an NVIDIA RTX TITAN GPU, and the batch size is set to 64. The average training time is set to 200 epochs. Adam optimizer is used to optimize the model until convergence. We set the hyperparameters to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. Our model training also supports the MindSpore framework. In addition to the two proposed RS pretrained backbones, ViT and Swin Transformer with supervised and self-supervised pretraining on ImageNet are also used. When fine-tuning for different datasets, we only adjust the neural unit number in the last linear layer to the corresponding number of categories. The proportions of training samples in NWPU-RESISC are set to TR = 10% and TR = 20%, respectively. The two training ratios for the AID dataset are TR = 20% and TR = 50%, respectively. The training rate for the UCM dataset is set to TR = 80%. All the experiments are conducted five times. In the experiment, overall accuracy (OA) is used as the performance metric, which is the most commonly used in scene classification.

3) Comparison With the SOTA: Table II demonstrates the results of various approaches using the above foundation model backbone, compared with the SOTA algorithms. "ISP," "ISSP," and "RingMo" stand for "ImageNet supervised pretraining," "ImageNet self-supervised pretraining," and "RS self-supervised pretraining," respectively. The experimental results can be briefly split into three groups, of which the first group represents previous SOTA methods. In addition, the

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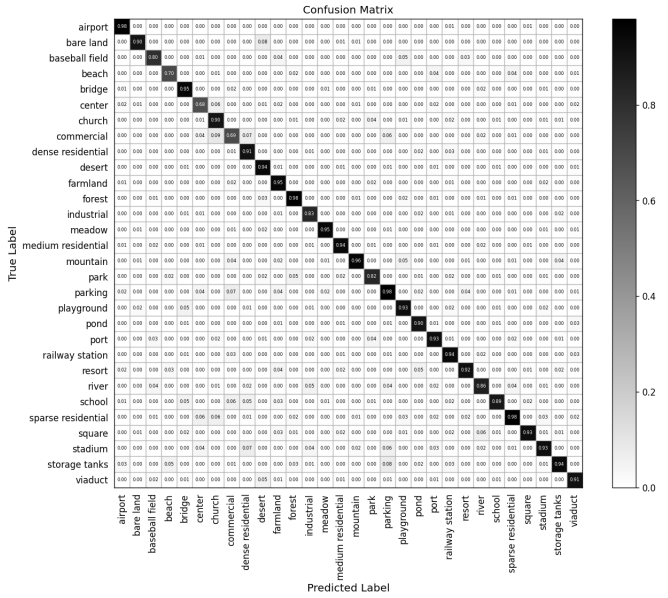

Fig. 6. Classification confusion matrix for RingMo (Swin-150W-200E) on 50% AID training data.

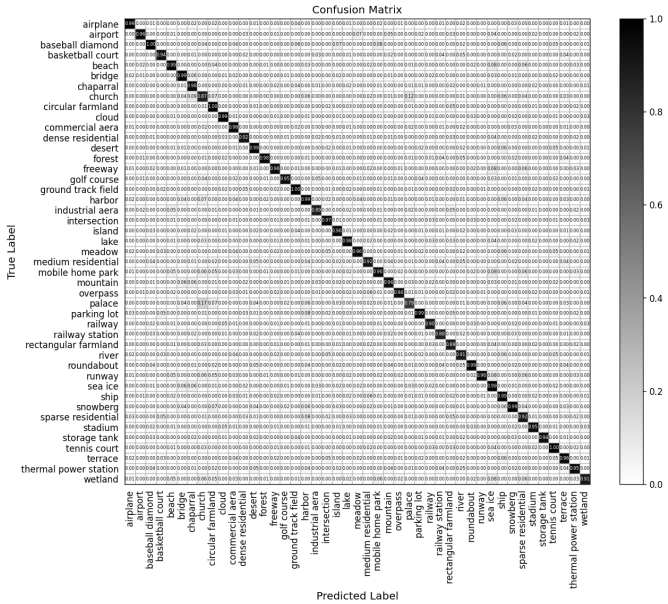

Fig. 7. Classification confusion matrix for RingMo (Swin-150W-200E) on 20% NWPU-RESISC45 training data.

ViT and Swin Transformer networks constitute two groups of experiments, respectively, and their foundation model weights are obtained by ISP, ISSP, and RingMo. We also consider the weights for different epochs and different amounts of RS data. The bold font indicates the best result in each group, and "*" indicates the best result among all the methods.

a) AID: As shown in Table II, the proposed RingMo can assist the ViT model to obtain higher accuracy. This demonstrates the effectiveness of RingMo, which uses massive RS data to provide better initial points for downstream fine-tuning, which suits scene classification tasks at different resolutions. It can be observed from Table II that the RingMo (Swin-200W-200E) method has the best results, followed by RingMo (Swin-150W-200E†), and the classification results obtained by RingMo (Swin-100W-200E) are relatively worse. However, all the RingMo results are close to or even better than the performance of some previous SOTA methods, which shows the superiority of our approach. In addition, the classification confusion matrix is provided to show the prediction results of the RSSP-Swin-B-150W-200E method (see Fig. 6). We note that the proposed RingMo achieves excellent performance in most classes.

b) NWPU-RESISC45: To further demonstrate the effectiveness of RingMo, the NWPU-RESISC45 dataset is used. The results in Table II show that RingMo can help the model achieve higher accuracy, even surpassing the existing SOTA methods when TR = 20%. This proves that the raw information of massive RS data shows a certain improvement on the NWPU-RESISC45 benchmark. Furthermore, Fig. 7 demonstrates the classification confusion matrix, to show the prediction results of the proposed RingMo (Swin-150W-200E) among different classes. It can be observed that compared with pretraining on ImageNet, the proposed RingMo can provide training initial values that are more practical for scene classification.

c) UCM: After the first two experiments on scene classification, to further evaluate the effect of our approach, the classic UCM is used. We observe that the proposed RingMo brings significant performance gains. Compared with ISSP (ViT), the proposed RingMo (ViT-150W-200E) improves OA by 0.43%. For the more advanced computational model Swin Transformer, the increase in accuracy is also very significant. Furthermore, to further demonstrate the remarkable effect of RingMo, it is compared with dedicated methods in RS. The results show that RingMo can help vision models to better understand RS scenes, which outperforms numerous dedicated models, as shown in Table II. This proves that even for deep networks designed for natural image classification, RingMo can be used for RS scene classification tasks without changing any network structure. Therefore, we think that RingMo enhances the generalization capability of vision methods and can be directly applied to RS data. The classification confusion matrix of RingMo (Swin-150W-200E) is shown in Fig. 8, and it achieves excellent performance on almost all the scene categories.

4) Ablation Study: Number of foundation model training epochs. Table II also demonstrates that foundation models with more epochs have the better representational capability. The performance of RingMo (Swin-150W-200E) is significantly better than that of RingMo (Swin-150W-100E), which proves that the generalization ability of the model is better with a longer pretraining time. Because of the huge amount of data in pretraining, short-term SSL is difficult to guarantee that efficient parameters are learned. The increase in pretraining time means that the model's understanding of large-scale RS data is gradually deepened. Considering the cost of training time, we mainly use foundation models at 200 epochs to record performance in other settings.

Amount of data for training foundation model. We also verify the effect of pretraining data amount on RingMo

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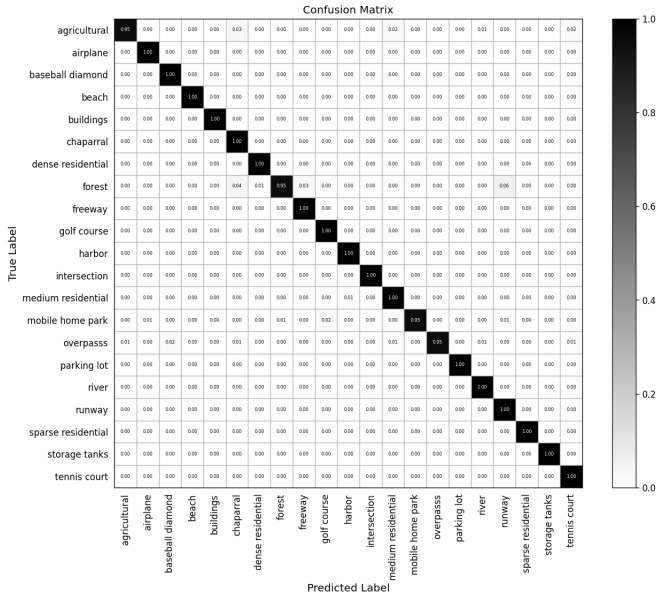

Fig. 8. Classification confusion matrix for RingMo (Swin-150W-200E) on 80% UCM training data.

performance. As shown in Table II, when the amount of pretraining data raises, the model accuracy on downstream fine-tuning increases accordingly. This suggests that raw information from massive RS data can improve downstream scene classification. The experimental results also show that RingMo can assist in fine-tuning convergence on scene classification tasks. Especially on small datasets, RingMo can achieve better performance as the amount of pretraining data raises. It enhances the understanding of RS scenes by processing richer samples. When there are many training samples, such as AID (TR = 50%) and UCM (TR = 80%), the performance of the deep model can be further explored. Whether ViT or Swin Transformer is used, RingMo shows superior performance in almost all the settings. Since manual annotation is not required, RingMo can theoretically use more RS data for pretraining to improve performance. For the convenience of analysis here, we only show three cases of 1 million, 1.5 million, and 2 million.

PIMask. The impact of the proposed PIMask is analyzed in this section. The results in Table II show that PIMask can further improve the upper limit of model performance under some settings, e.g., the accuracy improves OA by 0.25% on NWPU-RESISC45 (TR = 20%). Since PIMask does not completely mask each patch, the information of some small objects is preserved and SSL is facilitated. Therefore, for RS datasets containing a large number of small objects, the proposed PIMask can enhance the scene understanding and improve the classification performance.

C. RS Object Detection

Based on the optical camera carried by RS satellite, the ground objects are observed from a bird's eye view and usually have a regular rigid geometric structure. Therefore, these kinds of objects usually use the oriented bounding box (OBB) to derive more refined localization. There are other kinds of irregular geometric objects in RS scene, which is called component objects. Due to their irregular outline, the horizontal bounding box (HBB) is applied. It possesses a characteristic of "local invariance and global deformation," bringing more challenges for localization and recognition. We have carried out comparison and ablation experiments on the RS fine-grained category dataset (OBB) and dataset consisting of complex component objects (HBB), respectively.

1) Dataset Introduction: The experiments are carried out on RS fine-grained category dataset FAIR1M (oriented object detection task) and the DIOR dataset covering complex component objects (horizontal object detection task).

1) *FAIR1M [4]:* It contains more than 1 million instances. Over 15 000 images with a resolution of 0.3 to 0.8 m from various platforms are provided. The whole objects are annotated to five categories and 37 subcategories by OBBs. It includes five categories, which are airplane, vehicle, ship, road, and court. Each of these categories consists of several different fine-grained subcategories. We remove three other-* categories and compare the remaining 34 categories of models. In our experiments, we rename the above categories as C1–C34 in order.
2) *DIOR [5]:* DIOR is a large-scale, public RS dataset with 20 object categories. It contains 23 463 images and 190 288 instances in total. HBB annotation is provided for model training and testing. Besides the various RS scenes, it also possesses objects with different geometries. In the RS scene, there exist some objects covering a relatively large area and consisting of several regular components. Like the airport, expressway service area, Golf course, and train station in DIOR, they are complex composite objects more challenging to detect. In our experiments, we focus more on the results of the complex composite object categories.

2) Detailed Experimental Settings: We use the Bboxtoolkit in the OBBDetection framework [114], [115] to complete the FAIR1M datasets cropping. We crop the training set for 126 454 and test set 161 913 patches in the size of 512 × 512 in every 128 gaps. Data augmentation includes random horizontal, and vertical flipping is implemented. For the DIOR dataset, the images have already been cropped into 800 × 800 pixels. It is divided into a training set for 5862 patches, a validation set for 5863 patches, and a test set for 11 738 patches. During training, we mix the training set and validation set, and a randomly horizontal flipping is applied for data augmentation.

The models for two datasets adopt the AdamW optimizer. The LR is set to 0.0001, and its weight decay is 0.05. The training strategy of 36 epochs is adopted on both the datasets. We set the batch size to 16 for the FAIR1M dataset, and DIOR is 4. The LR adopts the warmup strategy in the first 500 iterations. The proportion is set to 0.001. It will reduce to one-tenth of the current value at the 24th and 33rd epochs. We conduct our experiments using an NVIDIA RTX TITAN GPU.

Based on the framework of OBBDetection [114], [115], we constructed our detection baseline model for the FAIR1M

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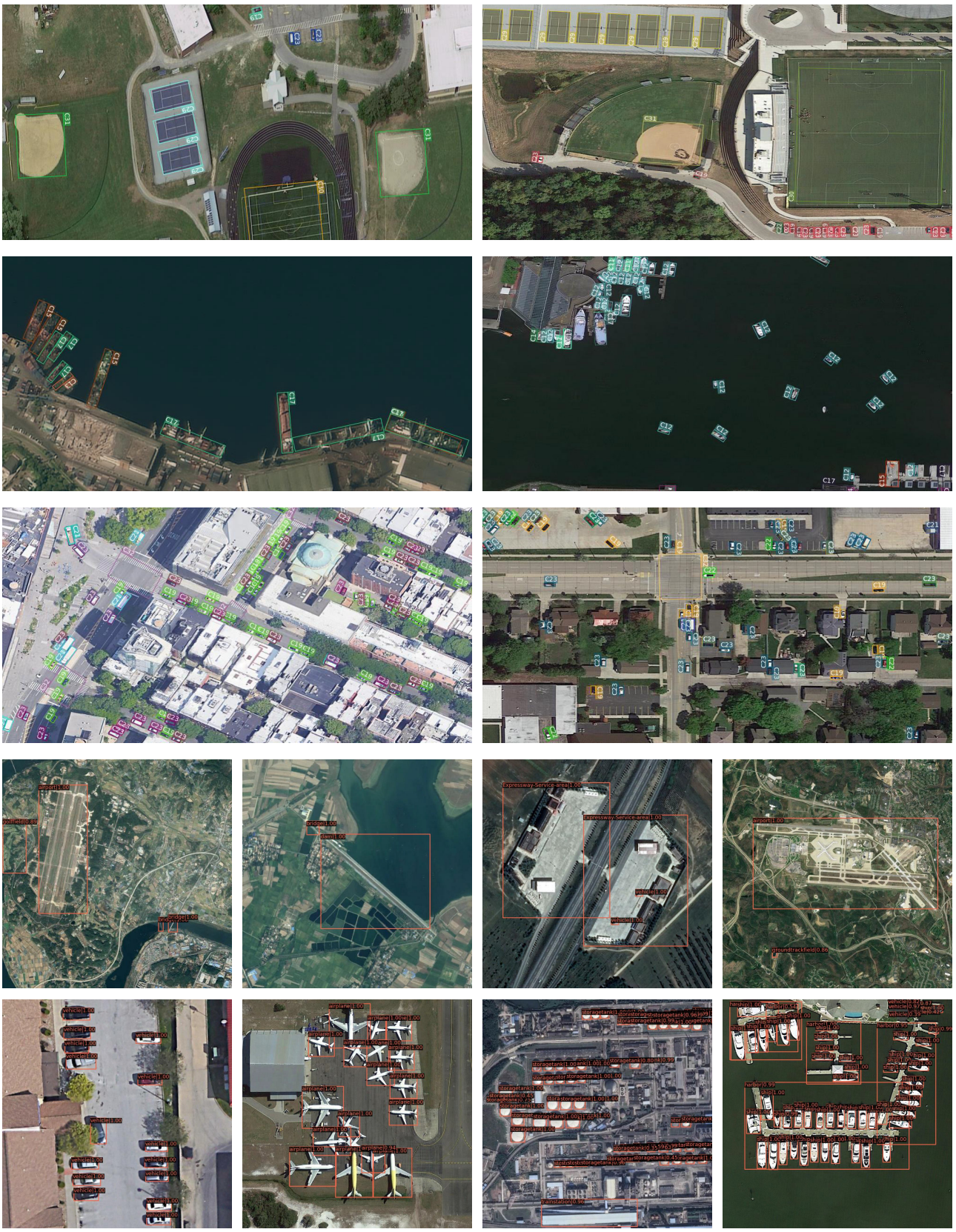

Fig. 9. Visualization results of the proposed model applying the RingMo (Swin-150W-200E†) method. Note that the first three rows are the results from the FAIR1M dataset, and the last three rows are from the DIOR dataset.

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TABLE III
COMPARISONS WITH THE SOTA MODELS ON THE FAIR1M DATASETS. THE UNIT OF DATA IN THE TABLE IS %

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	mAP
	C18	C19	C20	C21	C22	C23	C24	C25	C26	C27	C28	C29	C30	C31	C32	C33	C34	
Gliding Vertex [94]	35.43	47.88	15.67	48.32	0.01	40.11	39.31	16.54	16.56	0.01	9.12	23.34	1.23	15.67	15.43	15.32	25.43	29.92
	13.56	66.23	23.43	46.78	36.56	53.78	14.32	16.39	16.92	28.91	48.41	80.31	53.46	66.93	59.41	16.25	10.39	
RetinaNet [95]	38.46	55.36	24.75	51.84	0.81	40.5	41.06	18.02	19.94	1.7	9.57	22.55	1.33	16.37	19.11	14.26	24.7	30.67
	15.37	65.2	22.42	44.17	35.37	52.44	19.17	1.28	17.03	28.98	50.58	81.09	52.5	66.76	60.13*	17.41	12.58	
Cascade-RCNN [96]	40.42	52.86	29.07	52.47	0	44.37	38.35	26.55	17.54	0	12.1	28.84	0.71	15.35	18.53	14.63	25.15	31.18
	14.53	68.19	28.25	48.62	40.4	58	13.66	0.91	16.45	30.27	38.81	80.29	48.21	67.9	55.67	20.35	12.62	
Faster-RCNN [97]	36.43	50.68	22.5	51.86	0.01	47.81	43.83	17.66	19.95	0.13	9.81	28.78	1.77	17.65	16.47	16.19	27.06	32.12
	13.16	68.42	28.37	51.24	43.6	57.51	15.03	3.04	17.99	29.36	58.26	82.67	54.5	71.71	59.86	16.92	11.87	
RoI Transformer [98]	39.58	73.56	18.32	56.43	0	47.67	49.91	27.64	31.79	0	14.31	28.07	1.03	14.32	15.97	18.04	26.02	35.29
	12.97	68.8	37.41	53.96*	45.68	58.39	16.22	5.13	22.17	46.71	54.84	80.35	56.68	69.07	58.44	18.58	31.81	
OFA-Net [99]	43.20	87.58	23.58	57.78	30.71	54.31	67.83	73.65*	73.56	29.25	9.90	68.15*	11.70	30.37	11.83	26.80	38.68	43.73
	35.42	69.62	33.67	44.69	49.23	71.61*	13.45	6.06	17.36	5.35	49.20	83.75	61.29	88.77	57.97	24.68	35.82	
oriented-RCNN [100]	38.65	84.96	20.05	55.57	25.55	48.58	68.75	71.18	74.43*	32.85	20.46*	67.63	11.13	32.89*	16.22	26.45	39.03	44.30
	34.27	70.88*	45.01*	48.98	52.45	70.19	14.50	4.42	19.81	7.51	56.30*	80.41	68.13*	89.06*	57.92	21.87	30.11	
Ours																		
OBB-ISP(Swin)	43.97	**88.77***	38.97	58.29	50.88	54.09	65.14	59.19	65.94	**35.00***	**19.78**	56.77	26.84	26.31	**21.41***	**44.97***	**47.39***	45.91
	34.30	61.16	34.10	50.04	52.31	56.24	15.99	**7.43***	17.03	37.14	54.99	81.43	53.56	**87.66**	54.25	24.20	35.46	
OBB-ISSP(Swin)	**49.07***	88.48	37.43	**61.44***	53.47	53.10	**69.81***	56.81	**69.37**	33.81	13.56	**60.54**	25.09	29.46	17.59	45.05	42.99	45.76
	37.77	**61.58**	34.34	**50.28**	51.38	**56.57**	21.26	1.94	17.62	35.54	53.42	80.65	**58.39**	76.32	55.48	23.31	33.08	
OBB-RingMo(Swin-100W-200E)	46.70	88.24	38.18	56.53	**56.20***	53.77	67.52	57.23	67.71	32.46	16.63	55.71	26.64	30.18	15.84	42.18	43.29	45.82
	33.78	61.19	34.69	49.07	52.63	56.17	**21.95***	5.42	17.86	32.71	**56.18**	84.02	53.62	87.53	54.61	25.86	35.64	
OBB-RingMo(Swin-150W-100E)	46.82	87.82	37.28	56.95	**56.20***	53.67	64.69	56.08	63.54	34.69	19.14	55.79	28.76	29.30	16.90	42.22	42.73	45.86
	33.79	59.89	29.94	47.43	51.99	55.32	21.75	4.87	19.35	**39.69***	54.25	84.37	55.49	86.27	55.93	29.61	36.66	
OBB-RingMo(Swin-150W-200E)	47.04	87.96	37.24	57.02	55.74	53.62	64.85	58.01	64.56	34.53	19.24	55.96	**28.82***	28.96	16.20	41.73	42.92	46.02
	34.24	60.55	29.81	47.43	52.09	55.85	21.78	4.91	19.46	39.66	54.17	84.19	56.23	86.51	**56.10**	**30.73***	**36.57***	
OBB-RingMo(Swin-150W-200E)†	48.02	88.71	**39.19***	57.65	53.60	**54.40***	66.82	59.24	67.76	33.59	16.18	56.38	27.89	29.44	16.30	43.34	43.02	46.16
	35.02*	61.16	**35.25**	48.70	**52.94***	56.11	21.82	4.93	19.78	36.74	54.74	**84.46***	56.34	86.20	54.36	27.98	35.26	
OBB-RingMo(Swin-200W-200E)	48.12	88.51	39.12	57.65	53.91	54.01	66.56	**59.43**	67.12	33.76	16.18	56.41	28.55	**29.56**	16.24	43.22	42.98	**46.21***
	34.97	61.09	34.32	48.63	52.91	56.23	21.82	4.91	**19.82***	37.32	54.88	84.42	56.45	86.42	55.21	28.67	36.12	

*The correspondence between C1-C34 and object categories in the table (in order). C1: Boeing 737, C2: Boeing 747, C3: Boeing 777, C4: Boeing 787, C5: C919, C6: A220, C7: A321, C8: A330, C9: A350, C10: ARJ21, C11: passenger ship, C12: motorboat, C13: fishing boat, C14: tugboat, C15: engineering ship, C16: liquid cargo ship, C17: dry cargo ship, C18: warship, C19: small car, C20: bus, C21: cargo truck, C22: dump truck, C23: van, C24: trailer, C25: tractor, C26: excavator, C27: truck tractor, C28: basketball court, C29: tennis court, C30: football field, C31: baseball field, C32: intersection, C33: roundabout, C34: bridge

dataset, named OBB [98], [100]. The model generates the oriented region of interests (RoIs) in the first stage and further aligns the filtered boxes in the second stage. All the hyperparameters' setting remains the default values. We also apply the FPN [107] model as our baseline for the DIOR dataset.

We train the detection model on the training set. For the FAIR1M dataset, we test the models on the benchmark test set. For the DIOR dataset, the testing result is derived from the test dataset. During evaluation, we compare the average precision (AP) of each category. The mean average precision (mAP) is also considered.

3) Comparison With the SOTA: Tables III and IV show the RS object detection results after fine-tuning from the foundation model on two datasets, respectively.

a) FAIR1M: Applying the ISSP method, the mAP of the model is slightly lower than the ISP method. The reduction is 0.15% mAP. Specifically, we observe that on the hard distinguishable categories (A330, tractor, passenger ship) from the FAIR1M dataset, the ISSP model performs poorly. These results become much higher when applying the RingMo method, leading to the SOTA level.

b) DIOR: On the DIOR dataset, the method comparison of ISP, ISSP, and RingMo is similar to the results on FAIR1M. Specifically, the reduction is 1.2% mAP from ISSP to ISP, which is more pronounced than the FAIR1M dataset. We focus on the performance of the complex component object and the small-scale object categories in the DIOR dataset. We observed that the ISSP model performed poorly in complex composites (airports, golf courses, train stations) and small objects (aircraft, ships, vehicles) in the DIOR dataset. These categories recover under RingMo or even exceed the performance under ISP.

The experimental results of ISSP are inferior to ISP on both the datasets. We analyze that this may be due to the differences in the modeling of feature representations between supervised learning and generative SSL. Compared with supervised learning for image-level modeling, generative SSL achieves pixel-level modeling. Thus, a more general feature representation can be derived from the data. Therefore, also pretrained in different data domains (RS and natural scenes), ISSP amplifies the differences in data domains to a certain extent, which is inferior to the ISP method. Specifically, it is more pronounced in hard distinguishable, small-scale, and complex composite categories that require locality inductive bias and high-level semantic information. These results become much higher when applying the RingMo method, leading to the SOTA level. It can be concluded that whether it is locality inductive bias or high-level semantic information, the model has achieved significant improvement after applying RingMo.

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TABLE IV
COMPARISONS WITH THE SOTA MODELS ON DIOR DATASETS. THE UNIT OF DATA IN THE TABLE IS %

Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP
CV Methods																					
R-CNN [101]	35.6	43.0	53.8	62.3	15.6	53.7	33.7	50.2	33.5	50.1	49.3	39.5	30.9	9.1	60.8	18.0	54.0	36.1	9.1	16.4	37.7
RICNN [102]	39.1	61.0	60.1	66.3	25.3	63.3	41.1	51.7	36.6	55.9	58.9	43.5	39.0	9.1	61.1	19.1	63.5	46.1	11.4	31.5	44.2
RICAOD [103]	42.2	69.7	62.0	79.0	27.7	68.9	50.1	60.5	49.3	64.4	65.3	42.3	46.8	11.7	53.5	24.5	70.3	53.3	20.4	56.2	50.9
RIFD-CNN [104]	56.6	53.2	79.9	69.0	29.0	71.5	63.1	69.0	56.0	68.9	62.4	51.2	51.1	31.7	73.6	41.5	79.5	40.1	28.5	46.9	56.1
Faster R-CNN [97]	53.6	49.3	78.8	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	73.0	39.8	75.2	38.6	23.6	45.4	54.1
SSD [105]	59.5	72.7	72.4	75.7	29.7	65.8	56.6	63.5	53.1	65.3	68.6	49.4	48.1	59.2	61.0	46.6	76.3	55.1	27.4	65.7	58.6
YOLOv3 [106]	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	87.4	70.6	68.7	87.3	29.4	48.3	78.7	57.1
FPN-r50 [107]	54.1	71.4	63.3	81.0	42.6	72.5	57.5	68.7	62.1	73.1	76.5	42.8	56.0	71.8	57.0	53.5	81.2	53.0	43.1	80.9	63.1
FPN-r101 [107]	54.0	74.5	63.3	80.7	44.8	72.5	60.0	75.6	62.3	76.0	76.8	46.4	57.2	71.8	68.3	53.8	81.1	59.5	43.1	81.2	65.1
Mask-RCNN-r50 [108]	53.8	72.3	63.2	81.0	38.7	72.6	55.9	71.6	67.0	73.0	75.8	44.2	56.5	71.9	58.6	53.6	81.1	54.0	43.1	81.1	63.5
Mask-RCNN-r101 [108]	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	62.3	43.0	81.0	65.2
RetinaNet-r50 [95]	53.7	77.3	69.0	81.3	44.1	72.3	62.5	76.2	66.0	77.7	74.2	50.7	59.6	71.2	69.3	44.8	81.3	54.2	45.1	83.4	65.7
RetinaNet-r101 [95]	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
PANet-r50 [109]	61.9	70.4	71.0	80.4	38.9	72.5	56.6	68.4	60.0	69.0	74.6	41.6	55.8	71.7	72.9	62.3	81.2	54.6	48.2	86.7	63.8
PANet-r101 [109]	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1
CornerNet [110]	58.8	84.2	72.0	80.8	46.4	75.3	64.3	81.6	76.3	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9
RS Methods																					
CBD-E [111]	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	79.3	79.5	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1	67.8
CSFF [112]	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
SCRDet++(RetinaNet) [113]	64.3	79.0	73.2	85.7	45.8	76.0	68.4	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4
SCRDet++(FPN) [113]	66.4*	83.4	74.3	87.3	52.5	78.0	70.1	84.2	78.0	80.7	81.3	56.8	63.7	73.3	71.9	71.2*	83.4	62.3	55.6*	90.0*	73.2
Ours																					
FPN-ISP(Swin)	60.7	89.3	74.3	**87.8***	48.7	78.2	74.0	87.8	79.7	84.3	**82.6***	62.9	63.8	87.6	75.4	64.9	86.1	**73.3***	43.6	88.2	74.7
FPN-ISSP(Swin)	56.3	88.8	73.1	**87.8***	51.7	80.1	65.1	88.2	76.8	80.0	81.7	62.7	62.9	87.6	77.8	65.8	86.5	65.9	43.7	88.4	73.5
FPN-RingMo(Swin-100W-200E)	65.4	88.2	75.4	87.0	52.7	**81.1***	71.4	90.3	80.2	82.6	81.3	64.5	64.5	88.5	76.5	67.1	87.4	69.7	45.0	88.2	75.4
FPN-RingMo(Swin-150W-100E)	63.0	89.7	76.9	87.1	51.6	79.1	72.4	87.8	76.6	81.9	82.2	63.8	64.8	87.7	**78.8***	66.2	87.4	67.9	43.8	86.2	74.7
FPN-RingMo(Swin-150W-200E)	66.2	90.0	78.6	86.0	**53.4***	80.1	73.2	89.9	80.2	81.8	81.1	63.5	65.2	87.4	76.9	68.3	86.4	69.7	44.5	**88.7**	75.6
FPN-RingMo(Swin-150W-200E†)	64.8	**91.0***	77.8	86.4	53.0	78.7	**74.5***	**91.5***	80.3	83.8	80.4	64.6	65.9	**88.7***	74.2	67.1	87.6	71.1	**45.1**	87.6	75.7
FPN-RingMo(Swin-200W-200E)	**65.7**	87.7	**79.3***	87.2	52.5	80.5	73.1	91.0	**80.8***	**84.5***	80.8	**64.8***	**66.1***	**88.7***	75.9	**69.5**	**87.7***	70.0	44.9	88.2	**75.9***

*The correspondence between the abbreviations in the table and the categories. C1: Airplane, C2: Airport, C3: Baseball Field, C4: Basketball Court, C5: Bridge, C6: Chimney, C7: Dam, C8: Expressway Service Area, C9: Expressway Toll Station, C10: Golf Course, C11: Ground Track Field, C12: Harbor, C13: Overpass, C14: Ship, C15: Stadium, C16: Storage Tank, C17: Tennis Court, C18: Train Station, C19: Vehicle, C20: Wind Mill

The visualization results in Fig. 9 strongly show the adaptability in the RS scene and the powerful detection ability of the proposed model applying the RingMo method.

4) Ablation Study: We also complete a series of ablation studies on the proposed RingMo method. First, we compare our models fine-tuned from different amounts of data. As Tables III and IV show, the more data are added for training, the better performance the model reaches. Second, the foundation model pretrained from the 100 epochs versus 200 epochs is used for fine-tuning this task. We derive better results from longer pretraining epochs. Besides, the experimental results demonstrate that the model applying the proposed masking strategy performs better. To sum up, we draw several conclusions after analyzing the above results. 1) Our RingMo method benefits from having more training data, demonstrating the powerful data utilization capability of self-supervised methods. 2) General RS representation capabilities can be enhanced with relatively longer training iterations. 3) RS specific designs can further improve the performance, especially in detecting the extreme scale objects in RS scenes.

D. RS Semantic Segmentation

Semantic segmentation is viewed as a pixel-level classification task. We evaluate several of the models mentioned above on the task of RS semantic segmentation, which includes object and scene parsing segmentation. Object segmentation lays particular emphasis on the segmentation of foreground objects. Scene parsing aims to complete the pixel-level segmentation for the entire image.

1) Dataset Introduction: To prove the validity and generalization of the proposed models, we conduct experiments on the iSAID [37] and ISPRS Potsdam datasets which correspond to the above two segmentation subtasks, respectively.

1) *iSAID [37]:* The iSAID dataset is a semantic segmentation dataset of DOTA and contains 2806 aerial images with resolutions ranging from 800 × 800 to 4000 × 13 000 pixels. This dataset is mainly collected from Google Earth, and some images are captured by JL-1 satellite while the others are caught by GF-2 satellite. The whole data attach 655 451 instances for 15 object categories, and the remaining nonobject pixels are assigned as background. Therefore, this dataset has 16 categories in total. Among the 2806 high-resolution images, 1411, 458, and 957 images of them are separated as the training set, validation set, and test set, respectively. Since the annotations of the test set are unavailable, the validation set is used for testing in this article.
2) *ISPRS Potsdam:* The ISPRS Potsdam dataset published by ISPRS Commission WG $II/4$ consists of 38 high-resolution aerial images with a resolution of 0.5 m. The size of these images is 6000 × 6000 pixels, and there are 24 of them for training and 14 for testing. The dataset is annotated with six categories, including impervious surfaces, building, low vegetation, tree, car, and clutter.

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TABLE V
COMPARISONS WITH THE SOTA SEMANTIC SEGMENTATION METHODS ON THE VALIDATION SET OF iSAID

Method	IoU of each category (%)															mIoU (%)
	BD	BC	Bridge	GTF	Harbor	HC	LV	Plane	RA	Ship	SV	SBF	ST	SP	TC	
CV Methods																
FCN [116]	26.4	30.2	8.2	27.9	42.0	0	49.3	62.9	51.9	51.7	37.0	52.1	22.9	30.7	74.8	41.7
UNet [117]	36.5	22.9	7.5	5.5	45.6	0	49.9	74.7	46.5	49.0	35.6	9.7	0	38.0	78.6	39.2
DenseASPP [118]	67.5	56.6	29.6	52.3	51.1	0	57.1	78.1	64.8	61.1	38.4	74.1	50.0	43.3	86.1	56.8
Semantic FPN [119]	71.8	57.8	34.0	51.6	51.3	0	59.2	80.8	68.7	63.7	45.1	73.6	59.5	46.4	86.6	59.3
RefineNet [120]	72.3	61.1	32.6	52.8	51.1	23.0	58.2	79.9	65.6	63.8	42.4	74.4	58.6	43.4	85.3	60.2
PSPNet [121]	75.7	61.1	32.5	60.2	54.3	10.9	58.0	79.5	68.6	65.2	43.0	71.9	52.1	46.8	85.6	60.3
PointRend [122]	73.7	55.4	38.5	61.1	54.0	29.8	62.3	80.7	66.0	64.4	48.1	72.7	69.9	45.0	82.9	62.8
DeeplabV3+ [123]	72.8	56.5	32.2	58.9	52.6	31.4	59.1	79.8	67.7	63.9	42.9	72.9	52.5	46.1	84.9	60.8
EMANet [124]	66.2	56.0	42.1	18.8	47.5	33.4	58.2	78.5	46.9	63.1	41.0	46.4	68.4	38.9	82.7	55.4
GSCNN [125]	72.6	56.1	40.7	58.4	54.4	33.8	63.8	63.8	58.5	65.9	51.1	72.5	71.2	48.8	85.5	63.4
DANet [126]	71.4	50.9	28.6	52.5	48.8	30.4	57.5	80.9	40.6	60.2	42.1	63.3	63.0	46.1	84.7	57.5
CCNet [127]	68.9	57.1	34.0	56.8	45.9	31.6	57.6	75.8	57.2	61.4	38.3	75.0	65.7	36.5	82.9	58.3
EncNet [128]	70.0	55.2	36.8	46.3	48.0	34.8	57.2	76.1	59.8	59.7	38.7	69.8	64.9	42.4	84.2	58.9
HRNet [129]	74.0	59.4	33.8	61.5	52.2	14.9	62.1	81.7	52.9	65.9	46.9	75.6	68.9	44.2	86.9	61.5
AlignSeg [130]	76.2	62.1	28.7	52.0	54.8	31.2	60.7	82.9	56.2	67.4	50.3	71.2	68.9	45.7	86.2	62.1
OCR [131]	73.6	63.4	33.1	47.7	53.3	30.4	61.4	83.3	59.5	67.8	49.6	72.8	70.7	48.4	87.9	62.6
UperNet [132]	68.1	57.3	39.2	52.5	52.2	34.3	61.2	83.8	62.1	65.9	48.8	76.8	73.9*	44.5	70.7	61.9
RS Methods																
RANet [133]	72.5	53.2	45.3	47.1	55.6	38.1	60.1	83.1	70.5	67.1	49.3	58.8	61.3	41.8	85.1	62.1
DenseUNet [134]	76.1	57.7	33.9	49.5	54.6	0	54.7	82.2	65.9	66.1	46.2	71.9	50.4	45.1	86.2	58.7
HMANet [135]	74.7	60.5	29.0	54.6	51.9	32.6	59.7	83.8	62.9	65.4	50.3	70.2	70.9	51.4	88.7	62.6
GRDL [136]	78.5*	63.5	38.1	56.9	52.8	38.6	60.5	84.2	68.7	65.5	50.7	73.7	69.2	52.4*	88.7	65.1
FarSeg [137]	77.7	62.1	36.7	56.7	53.9	35.8	60.6	82.0	71.4	65.4	46.3	72.5	61.8	51.2	86.4	63.7
FactSeg [138]	78.4	64.9*	36.3	54.6	55.7	42.7*	62.7	84.1	69.4	68.3	49.5	73.6	56.8	51.5	88.9*	64.8
UperNet-SeCo [29]	66.9	54.5	38.9	45.9	51.4	33.2	58.2	83.3	52.3	63.9	44.8	71.6	71.7	9.3	69.9	57.2
UperNet(RSP-ViTAEv2-S) [22]	72.2	57.4	44.0	63.0	54.7	35.4	62.5	85.2	62.2	71.3	51.6	77.7	74.3*	47.0	70.4	64.3
Ours																
UperNet-ISP(Swin)	74.4	**64.2**	39.9	61.3	59.0	39.7	**64.4***	**85.7***	62.6	**75.7***	51.5	72.9	**73.6**	**49.1**	88.9	66.4
UperNet-ISSP(Swin)	75.3	58.6	39.4	55.5	54.0	38.2	62.2	84.9	65.2	68.5	**51.8***	71.5	73.2	45.3	86.4	64.3
UperNet-RingMo(Swin-100W-200E)	74.8	60.8	45.2	61.0	58.0	38.1	64.0	85.4	62.5	72.7	50.2	75.2	72.8	48.2	88.4	66.0
UperNet-RingMo(Swin-150W-100E)	75.2	61.1	41.3	58.1	56.4	34.5	63.0	85.1	66.7	70.9	48.6	74.7	73.4	48.1	87.3	65.2
UperNet-RingMo(Swin-150W-200E)	76.4	58.7	44.7	**63.5***	58.7	37.4	63.7	85.4	71.6	74.7	50.8	77.7	72.7	48.7	87.8	67.0
UperNet-RingMo(Swin-150W-200E†)	**77.3**	62.0	**46.0***	62.1	**59.4***	36.0	64.2	85.4	**73.1***	73.8	49.6	77.2	72.2	48.0	88.1	67.1
UperNet-RingMo(Swin-200W-200E)	77.0	62.5	43.2	63.0	58.9	**40.1**	63.9	**85.7***	67.3	73.5	51.2	**78.5***	73.0	48.9	**89.1**	**67.2***

*The 15 foreground categories of iSAID dataset: baseball diamond (BD), basketball court (BC) bridge, ground track field (GTF), harbor, helicopter (HC), large vehicle (LV), plane, roundabout (RA), ship, small vehicle (SV), soccer ball field (SBF), storage tank (ST), swimming pool (SP) and tennis court (TC)

In this section, images composed of near-infrared, red, and green spectral bands provided by the dataset are used for experiments.

2) Detailed Experimental Settings: The whole project of the segmentation downstream task is implemented based on mmsegmentation codebase [145] and we use the UperNet [132] as the segmentation framework. For optimization, the AdamW optimizer is used. The LR and weight decay are set to $6e^{-5}$ and 0.01, respectively. The polynomial decay rule is applied with a power of 1.0 and a minimal LR of 0. The linear warming-up strategy is adopted at the first 1500 iterations, with a ratio of $1e^{-6}$. To ensure the fairness of the results, all the models are trained 80k iterations with a batchsize of eight in the Potsdam dataset and 160k iterations with a batchsize of four in the iSAID dataset. Because the size of the images varies greatly, we crop the images of iSAID into patches of 896 × 896 pixels with an overlap of 384 (stride is 512), and the images of Potsdam are cropped into patches of 512 × 512 pixels with an overlap of 128 (stride is 384). For both the datasets used, we use some regular data augmentation strategies, including random horizontal flip, scaling (from 0.5 to 2.0), and random crop. Considering evaluation criteria commonly used in segmentation, we report the $F1$ score per category, mean $F1$ score (mF1), and OA metrics for the Potsdam dataset. Note that the clutter category is ignored during model training and testing. As for the iSAID dataset, we report the intersection over union (IoU) for each foreground class (15 in total). The mean IoU overall categories (16 in total, including background) are also provided.

3) Comparison With the SOTA:

a) iSAID: Table V shows the segmentation accuracy results of our approach and other SOTA methods on the iSAID dataset. No matter for the CV methods or the methods especially proposed for RS scenes, it can be observed that the results obtained using Swin Transformer as the backbone of the UperNet network are generally better than SOTA. In addition, from our series of results in Table V, it can be observed that our UperNet-RingMo (Swin-100W-200E) trained with 100w data gives slightly worse results compared with UperNet-ISP (Swin), but the results of the foundation model trained with 150w and 200w data are better than UperNet-ISP (Swin). This is because we use a generative self-supervised approach to pretrain the model with RS data, which can achieve better results when the amount of data increases and is consistent with the trend in the CV field. Compared with UperNet-ISSP (Swin), all our proposed foundation models have better performance.

b) ISPRS potsdam: Table VI shows the segmentation accuracy results of our approach and other SOTA methods on the Potsdam dataset. It is obvious that the UperNet using Swin Transformer as the backbone with our foundation model

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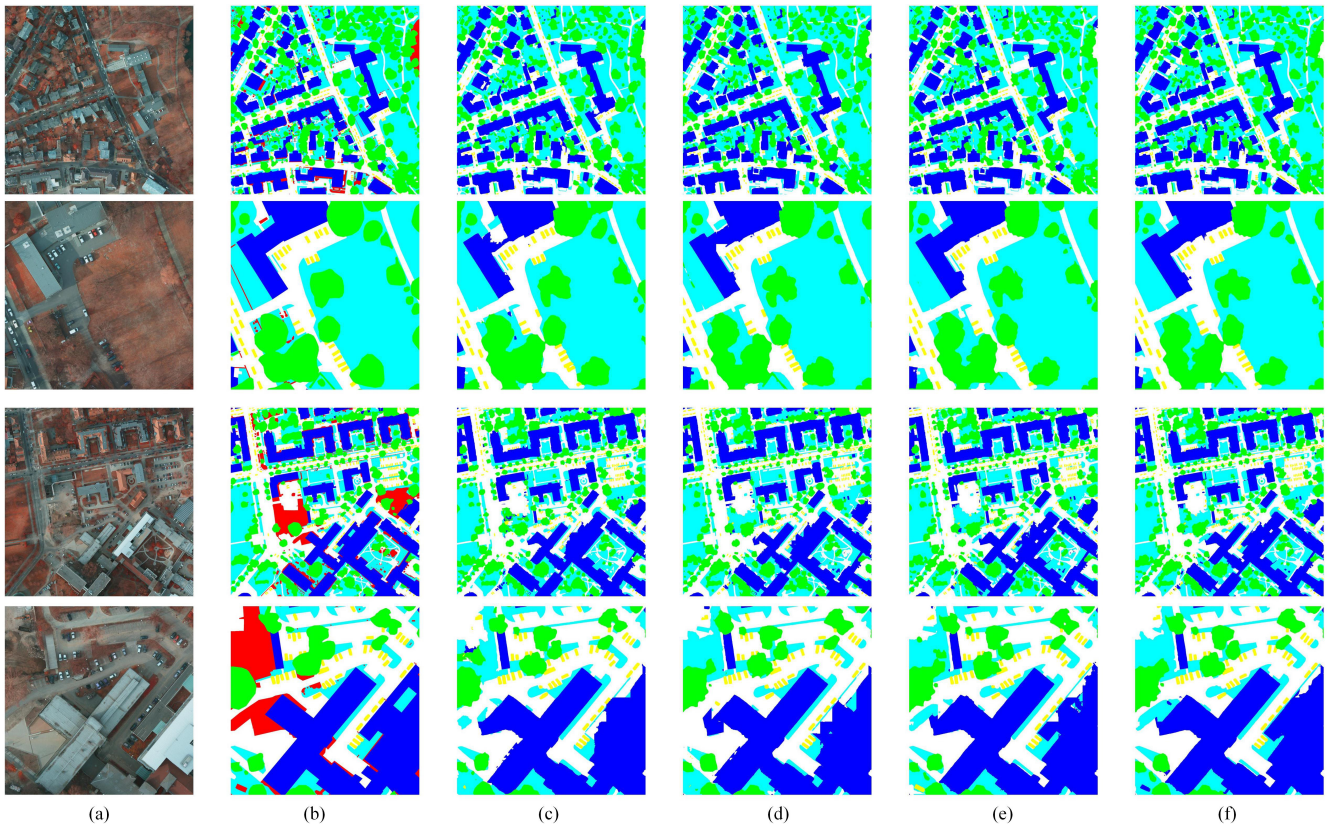
TABLE VI
COMPARISONS WITH THE SOTA SEMANTIC SEGMENTATION METHODS ON THE TEST SET OF POTSDAM

Method	F1 score per category (%)					mF1 (%)	OA (%)
	Imper. surf.	Building	Low veg.	Tree	Car		
CV methods							
FCN [139]	88.61	93.29	83.29	79.83	93.02	87.61	85.59
DANet [126]	91.61	96.44	86.11	88.04	83.54	89.14	89.72
PSPNet [121]	91.61	96.30	86.41	86.84	91.38	90.51	89.45
DeeplabV3+ [123]	92.35	96.77	85.22	86.79	93.58	90.94	89.74
UperNet [132]	93.14	96.75	86.30	86.13	91.56	90.78	91.26
RS methods							
S-RA-FCN [140]	91.33	94.70	86.81	83.47	94.52	90.17	88.59
UZ_1 [141]	89.30	95.40	81.80	80.50	86.50	86.70	85.80
UFMG_4 [142]	90.80	95.60	84.40	84.30	92.40	89.50	87.90
Multi-filter CNN [143]	90.94	96.98	76.32	73.37	88.55	85.23	90.65
HMANet [135]	92.38	96.08	86.93	88.21*	95.44*	91.81	90.46
LANet [144]	93.05	97.19*	87.30*	88.04	94.19	91.95*	90.84
UperNet-SeCo [29]	91.21	94.92	85.12	84.89	89.02	89.03	89.64
UperNet(RSP-ViTAEv2-S) [22]	93.05	96.62	86.62	85.89	91.01	90.64	91.21
Ours							
UperNet-ISP(Swin)	93.60	**97.15**	86.97	86.15	92.05	91.18	91.66
UperNet-ISSP(Swin)	93.15	96.62	86.56	86.10	91.61	90.81	91.27
UperNet-RingMo(Swin-100W-200E)	93.48	97.01	86.91	**86.36**	92.18	91.19	91.62
UperNet-RingMo(Swin-150W-100E)	93.47	97.06	86.97	86.28	92.00	91.15	91.62
UperNet-RingMo(Swin-150W-200E)	93.48	97.06	86.92	86.35	92.15	91.19	91.63
UperNet-RingMo(Swin-150W-200E†)	**93.63***	97.13	87.08	**86.36**	92.16	**91.27**	**91.74***
UperNet-RingMo(Swin-200W-200E)	93.54	97.11	**87.09**	86.32	**92.19**	91.25	91.70



(a) (b) (c) (d) (e) (f)

Fig. 10. Visualation results of our method and compared methods on the Potsdam dataset. (a) Images. (b) Labels. (c) UperNet-ISP (Swin). (d) UperNet-ISSP (Swin). (e) UperNet-RingMo (Swin-150W-200E). (f) UperNet-RingMo (Swin-150W-200E†). Our method achieves more accurate segmentation results compared with other pretrained models.

achieves higher OA metric compared with both CV methods and methods of RS field. The UperNet-ISP (Swin) achieves relatively high mIoU and OA accuracy compared with the UperNet-ISSP (Swin). Our foundation model better exploits

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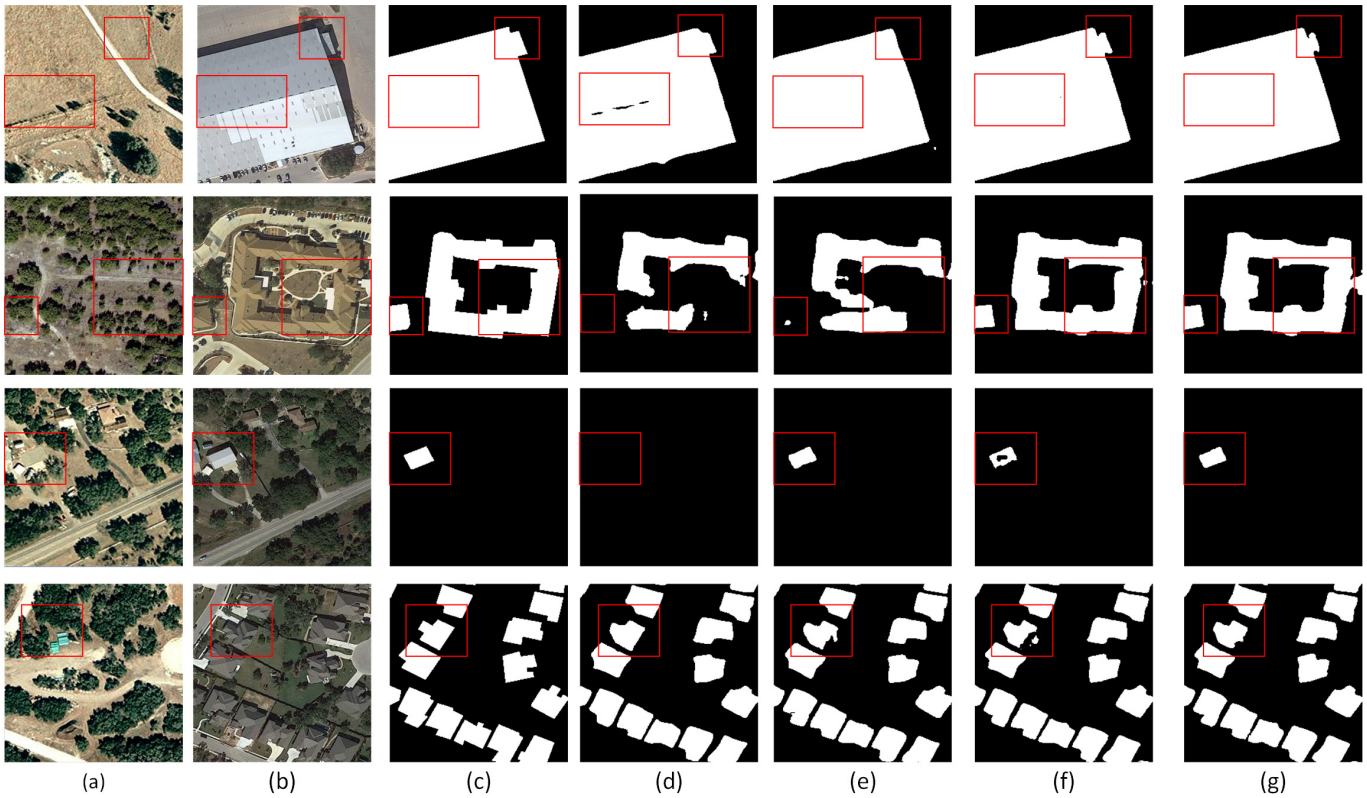

Fig. 11. Visualation results of our method and compared methods on the LEVIR-CD dataset. (a) T0 images. (b) T1 images. (c) Label. (d) BIT-ISP (Swin). (e) BIT-ISSP (Swin). (f) BIT-RingMo (Swin-150W-200E). (g) BIT-RingMo (Swin-150W-200E†). Our method achieves better performance compared with other models.

the characteristics of RS scenes to achieve better performance compared with them.

We visualize the segmentation results of our method and compared the methods on Potsdam data, as shown in Fig. 10. We show test results of two images with 6000 × 6000 pixels in the first and third rows. Then, we show their local details with 2000 × 2000 pixels in the second and fourth rows. It can be obviously noted that the segmentation of the RingMo method is more accurate for long-scale buildings compared with the ISP method and ISSP method. Our model has more consistent and efficient segmentation performance for object edges.

4) Ablation Study: Amount of data for training foundation model. From Tables V and VI, we can note that when the amount of loaded model data gradually increases (100w → 150w → 200w), the segmentation performance on the two datasets gradually improves, especially on the iSAID dataset (66.0% → 67.0% → 67.2%).

The number of foundation model training epochs. For the foundation model with the same amount of 150w data, on the two segmentation datasets, the result of the foundation model obtained by training 200 epochs is better than the result of training only 100 epochs. Specifically, mIoU increased by 1.8% on the iSAID dataset. On the Potsdam dataset, the mF1 metric improves by 0.04%.

PIMask. We analyze the performance impact of the proposed mask method. From Tables V and VI, we can find that using our proposed mask can bring performance improvement on the downstream task of semantic segmentation. Specifically, on the iSAID dataset, mIoU improves by 0.1%, and on the Potsdam dataset, the OA metric improves by 0.11%.

E. RS Change Detection

Change detection is an important task in the field of RS. The purpose of change detection is to mark the changed areas in the same location area at different times. It plays an important role in urban expansion, deforestation, and damage assessment.

1) Dataset Introduction: We carry out experiments on the LEVIR-CD [146] dataset to verify the effectiveness of our method.

1) *LEVIR-CD:* Dataset is a new large-scale RS binary change detection dataset. The bitemporal images in the LEVIR-CD dataset were collected in TX, USA, with capture times ranging from 2002 to 2018. The dataset is a very high resolution (VHR) RS image containing 637 pairs of 1024 × 1024 pixels. The labels of the images are binary. The LEVIR-CD dataset focuses on building growth and building decline.

2) Detailed Experimental Settings: Our model adopts BIT with Swin Transformer as the backbone. The model is trained with AdamW optimizer with a weight decay 0.01 and beta values equal to (0.9, 0.999). The LR is $6e^{-5}$ for the LEVIR-CD dataset and linearly decays to 0 until trained for 200 epochs. The batch size is 8. We use the sliding window strategy with

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TABLE VII
COMPARISONS WITH THE SOTA MODELS ON THE LEVIR-CD DATASET. WE USE PRECISION, RECALL, AND $F1$ VALUE AS THE EVALUATION METRICS. ALL THE VALUES IN THE TABLE DEFAULT TO PERCENT SIGNS

Method	LEVIR-CD		
	Precision	Recall	F1
FC-EF [147]	86.91	80.17	83.40
FC-Siam-Conc [147]	91.99	76.77	83.69
FC-Siam-Di [147]	89.53	83.31	86.31
CLNet [148]	89.80	90.30	90.00
SNUNet-CD [149]	89.18	87.17	88.16
IFN [150]	94.02*	82.93	88.13
DTCDSCN [151]	88.53	86.83	87.67
DASNet [152]	80.76	79.53	79.91
STANet [153]	83.81	91.00	87.34
BSFNet [154]	82.70	94.00*	88.00
HRTNet [155]	85.43	91.77	88.48
CDNet[156]	91.60	86.50	89.00
BIT-RES [85]	89.24	89.37	89.31
ChangeFormer [157]	92.05	88.80	90.40
CS-HSNet [15]	93.12	88.58	90.79
BiT-Seco-ResNet-50 [29]	–	–	90.14
BiT-RSP-VITAEv2-S [22]	–	–	90.93
BiT-ISP(Swin)	92.38	89.98	91.16
BiT-ISSP(Swin)	92.04	89.76	90.94
BIT-RingMo(Swin-100W-200E)	92.42	91.00	91.70
BIT-RingMo(Swin-150W-100E)	92.45	90.94	91.81
BIT-RingMo(Swin-150W-200E)	92.57	91.08	91.82
BIT-RingMo(Swin-150W-200E†)	**92.65**	91.08	**91.86***
BIT-RingMo(Swin-200W-200E)	92.47	**91.17**	91.85

an overlapping stride of 100 pixels to crop the images of the LEVIR-CD dataset into 256 × 256 pixels. We train the model for 200 epochs in all the datasets. The experiment setting of the baseline methods is similar to ours. We use the $F1$-score, precision, and, recall with regard to the change category as the main evaluation indices. All the experiments are implemented by the Pytorch framework [69] with NVIDIA RTX TITAN GPU.

3) Comparison With the SOTA: We compare our method with the SOTA methods. In terms of the results on the LEVIR-CD dataset, our method achieves optimal results on $F1$. Besides, precision and recall also perform well. Specifically, in RS scenarios, our proposed foundation model outperforms ISSP models in unsupervised training conditions on change detection. The optimal effect is achieved when using 200w RS data to train the foundation model. Transformer-based baseline methods, such as BIT and ChangeFormer, are suboptimal methods compared with other baseline methods, which demonstrate that Transformer-based methods are promising for change detection tasks. Our approach outperforms the above Transformer-based methods, which means our method is excellent.

We visualize some 256 × 256 patches of the LEVIR-CD dataset in Fig. 11. Intuitively, the method with our foundation model performs better for change segmentation than loading the ImageNet pretrained model. For small and slender objects, our change detection results are more refined, which proves that loading our RS pretrained model for training is more suitable for RS change detection tasks.

4) Ablation Experiments: The quantitative analysis results are shown in Table VII. First, foundation models trained with different data sizes (100w, 150w, 200w) have regular effects on the change detection results. More pretraining data further improve the performance. This result indicates that increasing the size of RS image data is more conducive to achieving excellent results in downstream tasks. Second, the foundation model trained with more epochs (100 epochs, 200 epochs) is more effective when applied to change detection tasks. Longer training time ensures that the pretrained model is better optimized, which is more beneficial for the model to be applied to the change detection task. Third, the improved mask method improves the performance of the change detection task. If the pixels inside the patch are not completely masked, it is beneficial to preserve the information of small objects in the RS datasets.

V. DISCUSSION

Based on the experimental comparison on the eight datasets of the four downstream tasks above, we analyze the experimental results systematically and comprehensively.

The result of ISSP is generally lower than that of the ISP method. We believe that this is mainly caused by the learning difference between the feature representation ability of the two methods. Compared with ISP, ISSP introduces more local inductive bias for model learning. In other words, its pixel-level modeling makes the model pay more attention to the general feature representation in the data. But the ISSP weights obtained in natural scenes may lead to more difficult adaption in RS due to the domain bias problem.

RingMo trained with RS data performs better than ISSP trained with natural scene images and even outperforms supervised trained ISP. It illustrates the effectiveness of RS images for the foundation model. In the ablation experiment corresponding to RingMo, the model performance can be further improved with the increase in the amount of data and the number of training iterations, including innovation introduction. It further shows that the effective use of more RS data can obtain more general feature representation, which helps the model gain performance in downstream tasks.

In addition, in the comparison of the four tasks, we observe the following results. RingMo is superior to ISP in tasks involving pixel-level modeling and object localization. For example, on datasets such as LEVIR-CD and DIOR, the RingMo model exceeds the ISP model by $0.54F1/0.7$mAP on average. In image-level classification tasks such as scene classification, the results of RingMo are similar to other supervised learning methods, even lower than them. We analyze that it is mainly due to the granularity difference between the RingMo and supervised learning methods. RingMo applies MIM to improve the performance of pixel-level feature encoding and decoding tasks. However, image-level classification adopted by supervised learning methods matches tasks such as scene classification. We will further design a foundation model for the issue that takes into account both the image-level and pixel-level tasks.

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VI. Conclusion

In this article, we propose an RS foundation model framework, which includes a large-scale dataset and a foundation model training method. First, a dataset of two million RS images is developed without human supervision. The images are captured from satellite and aerial platforms, covering the majority of the world's scenes and objects, helping to improve the adaptability of the foundation model to more RS scenes. In addition, for deriving general RS feature representation, we further propose the RingMo MIM method, which enhances the issue of dense and small objects being easily disregarded in complicated RS scenes. Using the trained foundation model to fine-tune downstream applications, we achieve SOTA performance on eight datasets for four tasks, demonstrating the effectiveness of our proposed framework.

In the future, we aim to collect a larger amount of data with richer modalities and expand our framework to more diverse downstream applications, making the foundation model obtain more general and essential RS feature representations. We hope that our work can inspire more RS researchers to introduce generative self-supervised methods into the RS field.

References

[1] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[2] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 9646–9660, Nov. 2021.

[3] X. Zhao *et al.*, "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[4] X. Sun *et al.*, "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[5] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[6] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[7] F. Kun, S. Xian, and Q. Xiaolan, "Multi-satellite integrated processing and analysis method under remote sensing big data," *Nat. Remote Sens.*, vol. 25, no. 3, pp. 691–707, 2021.

[8] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 99, pp. 3735–3756, Jun. 2020.

[9] X. Lu, X. Sun, W. Diao, Y. Feng, P. Wang, and K. Fu, "LIL: Lightweight incremental learning approach through feature transfer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[10] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[11] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021.

[12] Y. Feng, X. Sun, W. Diao, J. Li, and X. Gao, "Double similarity distillation for semantic image segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 5363–5376, 2021.

[13] Z. Yang, Z. Yan, X. Sun, W. Diao, Y. Yang, and X. Li, "Category correlation and adaptive knowledge distillation for compact cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[14] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[15] Q. Ke and P. Zhang, "CS-HSNet: A cross-Siamese change detection network based on hierarchical-split attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9987–10002, 2021.

[16] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[17] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2022, doi: 10.1109/TNNLS.2022.3149394.

[18] X. Sun, A. Shi, H. Huang, and H. Mayer, "BAS^{4}Net: Boundary-aware semi-supervised semantic segmentation network for very high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5398–5413, 2020.

[19] Z. Zhu *et al.*, "Diamondnet: Ship detection in remote sensing images by extracting and clustering keypoints in a diamond," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. V-2-2020, pp. 625–632, Aug. 2020.

[20] Z. Zhu, X. Sun, W. Diao, K. Chen, G. Xu, and K. Fu, "AOPDet: Automatic organized points detector for precisely localizing objects in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[21] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[22] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, early access, May 25, 2022, doi: 10.1109/TGRS.2022.3176603.

[23] W. Li, H. Chen, and Z. Shi, "Semantic segmentation of remote sensing images with self-supervised multitask representation learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6438–6450, 2021.

[24] W. Li, K. Chen, H. Chen, and Z. Shi, "Geographical knowledge-driven representation learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[25] K. Ayush *et al.*, "Geography-aware self-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10181–10190.

[26] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[27] T. Chen, S. Kornblith, K. Swersky, M. Norouzi, and G. E. Hinton, "Big self-supervised models are strong semi-supervised learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 22243–22255.

[28] P. Akiva, M. Purri, and M. Leotta, "Self-supervised material and texture representation learning for remote sensing tasks," 2021, *arXiv:2112.01715*.

[29] O. Manas, A. Lacoste, X. Giro-i-Nieto, D. Vazquez, and P. Rodriguez, "Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9414–9423.

[30] K. Heidler *et al.*, "Self-supervised audiovisual representation learning for remote sensing data," 2021, *arXiv:2108.00688*.

[31] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," 2021, *arXiv:2111.06377*.

[32] Z. Xie *et al.*, "SimMIM: A simple framework for masked image modeling," 2021, *arXiv:2111.09886*.

[33] T.-Y. Lin *et al.*, "Microsoft coco: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.

[34] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst. (GIS)*, 2010, pp. 270–279.

[35] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[36] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[37] S. Waqas Zamir *et al.*, "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 28–37.

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[38] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[39] P. Jain, B. Schoen-Phelan, and R. Ross, "Multi-modal self-supervised representation learning for Earth observation," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3241–3244.

[40] Y. Long *et al.*, "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.

[41] G. Christie, N. Fendley, J. Wilson, and R. Mukherjee, "Functional map of the world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6172–6180.

[42] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[43] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.

[44] X. Chen, S. Xie, and K. He, "An empirical study of training self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9640–9649.

[45] Y. Tian, D. Krishnan, and P. Isola, "Contrastive multiview coding," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 776–794.

[46] X. Zheng, B. Kellenberger, R. Gong, I. Hajnsek, and D. Tuia, "Self-supervised pretraining and controlled augmentation improve rare wildlife recognition in UAV images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 732–741.

[47] V. Stojnic and V. Risojevic, "Self-supervised learning of remote sensing scene representations using contrastive multiview coding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1182–1191.

[48] K. Cha, J. Seo, and Y. Choi, "Contrastive multiview coding with electro-optics for SAR semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] H. Jung, Y. Oh, S. Jeong, C. Lee, and T. Jeon, "Contrastive self-supervised learning with smoothed representation for remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[50] H. Dong, W. Ma, Y. Wu, J. Zhang, and L. Jiao, "Self-supervised representation learning for remote sensing image change detection based on temporal prediction," *Remote Sens.*, vol. 12, no. 11, p. 1868, Jun. 2020.

[51] H. Chen, Y. Zao, L. Liu, S. Chen, and Z. Shi, "Semantic decoupled representation learning for remote sensing image change detection," 2022, *arXiv:2201.05778*.

[52] A. Van Oord, N. Kalchbrenner, and K. Kavukcuoglu, "Pixel recurrent neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1747–1756.

[53] A. Van den Oord, N. Kalchbrenner, L. Espeholt, O. Vinyals, and A. Graves, "Conditional image generation with PixelCNN decoders," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 4797–4805.

[54] L. Dinh, D. Krueger, and Y. Bengio, "NICE: Non-linear independent components estimation," 2014, *arXiv:1410.8516*.

[55] L. Dinh, J. Sohl-Dickstein, and S. Bengio, "Density estimation using real NVP," 2016, *arXiv:1605.08803*.

[56] D. P. Kingma and P. Dhariwal, "Glow: Generative flow with invertible 1×1 convolutions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 10236–10245.

[57] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.

[58] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, P.-A. Manzagol, and L. Bottou, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *J. Mach. Learn. Res.*, vol. 11, no. 12, pp. 1–38, 2010.

[59] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, and A. A. Efros, "Context encoders: Feature learning by inpainting," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2536–2544.

[60] M. Chen *et al.*, "Generative pretraining from pixels," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1691–1703.

[61] R. Zhang, P. Isola, and A. A. Efros, "Colorful image colorization," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 649–666.

[62] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[63] H. Bao, L. Dong, and F. Wei, "BEiT: BERT pre-training of image transformers," 2021, *arXiv:2106.08254*.

[64] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[65] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[66] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[67] T. Xiao, P. Dollar, M. Singh, E. Mintun, T. Darrell, and R. Girshick, "Early convolutions help transformers see better," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 30392–30400.

[68] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[69] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8026–8037.

[70] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[71] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.

[72] N. He, L. Fang, S. Li, A. Plaza, and J. Plaza, "Remote sensing scene classification using multilayer stacked covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6899–6910, Dec. 2018.

[73] Y. Yu and F. Liu, "A two-stream deep fusion framework for high-resolution aerial scene classification," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–13, Jan. 2018.

[74] Z. Chen, S. Wang, X. Hou, L. Shao, and A. Dhabi, "Recurrent transformer network for remote sensing scene categorisation," in *Proc. BMVC*, vol. 266, 2018, pp. 1–11.

[75] D. Zeng, S. Chen, B. Chen, and S. Li, "Improving remote sensing scene classification by integrating global-context and local-object features," *Remote Sens.*, vol. 10, no. 5, p. 734, May 2018.

[76] R. Minetto, M. P. Segundo, and S. Sarkar, "Hydra: An ensemble of convolutional neural networks for geospatial land classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6530–6541, Sep. 2019.

[77] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, 2019.

[78] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2018.

[79] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[80] F. Li, R. Feng, W. Han, and L. Wang, "High-resolution remote sensing image scene classification via key filter bank based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8077–8092, Nov. 2020.

[81] H. Sun, S. Li, X. Zheng, and X. Lu, "Remote sensing scene classification by gated bidirectional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 82–96, Jan. 2020.

[82] S. Wang, Y. Guan, and L. Shao, "Multi-granularity canonical appearance pooling for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 29, pp. 5396–5407, 2020.

[83] Z. Zhao, J. Li, Z. Luo, J. Li, and C. Chen, "Remote sensing image scene classification based on an enhanced attention module," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1926–1930, Nov. 2020.

[84] X. Zhang, W. An, J. Sun, H. Wu, W. Zhang, and Y. Du, "Best representation branch model for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9768–9780, 2021.

[85] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[86] B. Li, Y. Guo, J. Yang, L. Wang, Y. Wang, and W. An, "Gated recurrent multiattention network for VHR remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

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[87] S. Wang, Y. Ren, G. Parr, Y. Guan, and L. Shao, "Invariant deep compressible covariance pooling for aerial scene categorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6549–6561, Aug. 2021.

[88] G. Zhang *et al.*, "A multiscale attention network for remote sensing scene images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9530–9545, 2021.

[89] P. Deng, K. Xu, and H. Huang, "When CNNs meet vision transformer: A joint framework for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[90] Q. Bi, K. Qin, H. Zhang, and G.-S. Xia, "Local semantic enhanced ConvNet for aerial scene recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 6498–6511, 2021.

[91] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 15, 2021, doi: 10.1109/TNNLS.2021.3071369.

[92] Q. Zhao, S. Lyu, Y. Li, Y. Ma, and L. Chen, "MGML: Multigranularity multilevel feature ensemble network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 1, 2021, doi: 10.1109/TNNLS.2021.3106391.

[93] Q. Zhao, Y. Ma, S. Lyu, and L. Chen, "Embedded self-distillation in compact multi-branch ensemble network for remote sensing scene classification," 2021, *arXiv:2104.00222*.

[94] Y. Xu *et al.*, "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[95] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[96] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[97] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[98] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[99] Q. Ming, J. Song, and Z. Zhou, "Oriented feature alignment for fine-grained object recognition in high-resolution satellite imagery," 2021, *arXiv:2110.06628*.

[100] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[101] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[102] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[103] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[104] G. Cheng, J. Han, P. Zhou, and D. Xu, "Learning rotation-invariant and Fisher discriminative convolutional neural networks for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 265–278, Jan. 2019.

[105] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[106] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[107] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[108] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2961–2969.

[109] L. Cheng, X. Liu, L. Li, L. Jiao, and X. Tang, "Deep adaptive proposal network for object detection in optical remote sensing images," 2018, *arXiv:1807.07327*.

[110] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 734–750.

[111] J. Zhang, C. Xie, X. Xu, Z. Shi, and B. Pan, "A contextual bidirectional enhancement method for remote sensing image object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4518–4531, 2020.

[112] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 431–435, Mar. 2021.

[113] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Apr. 12, 2022, doi: 10.1109/TPAMI.2022.3166956.

[114] K. Chen *et al.*, "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.

[115] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[116] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[117] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Cham, Switzerland: Springer, 2015, pp. 234–241.

[118] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.

[119] A. Kirillov, R. Girshick, K. He, and P. Dollar, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6399–6408.

[120] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1925–1934.

[121] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[122] A. Kirillov, Y. Wu, K. He, and R. Girshick, "PointRend: Image segmentation as rendering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9799–9808.

[123] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[124] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9167–9176.

[125] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5229–5238.

[126] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[127] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[128] H. Zhang *et al.*, "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.

[129] K. Sun *et al.*, "High-resolution representations for labeling pixels and regions," 2019, *arXiv:1904.04514*.

[130] Z. Huang, Y. Wei, X. Wang, W. Liu, T. S. Huang, and H. Shi, "AlignSeg: Feature-aligned segmentation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 550–557, Jan. 2022.

[131] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 173–190.

[132] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 418–434.

[133] L. Mou, Y. Hua, and X. X. Zhu, "A relation-augmented fully convolutional network for semantic segmentation in aerial scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12416–12425.

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[134] R. Dong, X. Pan, and F. Li, "DenseU-Net-based semantic segmentation of small objects in urban remote sensing images," *IEEE Access*, vol. 7, pp. 65347–65356, 2019.
[135] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.
[136] R. Niu, X. Sun, Y. Tian, W. Diao, Y. Feng, and K. Fu, "Improving semantic segmentation in aerial imagery via graph reasoning and disentangled learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.
[137] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.
[138] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[139] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[140] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.
[141] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2016.
[142] K. Nogueira, M. Dalla Mura, J. Chanussot, W. R. Schwartz, and J. A. dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, 2019.
[143] Y. Sun, X. Zhang, Q. Xin, and J. Huang, "Developing a multi-filter convolutional neural network for semantic segmentation using high-resolution aerial imagery and LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 143, pp. 3–14, Sep. 2018.
[144] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.
[145] MMSegmentation Contributors. (2020). *MMSegmentation: OpenMMLab Semantic Segmentation Toolbox and Benchmark*. [Online]. Available: https://github.com/open-mmlab/mmsegmentation
[146] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.
[147] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[148] Z. Zheng, Y. Wan, Y. Zhang, S. Xiang, D. Peng, and B. Zhang, "CLNet: Cross-layer convolutional neural network for change detection in optical remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 247–267, May 2021.
[149] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[150] C. Zhang *et al.*, "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
[151] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.
[152] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.
[153] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[154] H. Du *et al.*, "Bilateral semantic fusion Siamese network for change detection from multitemporal optical remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[155] X. Hou, Y. Bai, Y. Li, C. Shang, and Q. Shen, "High-resolution triplet network with dynamic multiscale feature for change detection on satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 103–115, Jul. 2021.
[156] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[157] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.

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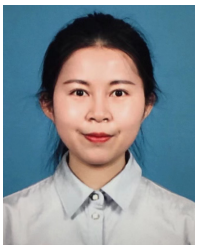

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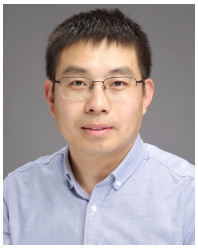

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